

Notes: Gravitation and Einstein's Equation

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Chapter 1

Gravitation

1.1 *Physics in Curved Spacetime*

Having established the mathematical framework of differential geometry, we are now prepared to examine the physics of gravitation as described by General Relativity. The subject fundamentally divides into two distinct but interconnected phenomena: how the gravitational field (spacetime curvature) influences the behavior of matter, and conversely, how matter determines the gravitational field.

In Newtonian gravity, these two elements are governed by two specific equations. First, the response of a body to a gravitational potential Φ is given by its acceleration:

$$\mathbf{a} = -\nabla\Phi \quad \implies \quad \frac{d^2x^i}{dt^2} = -\partial_i\Phi. \quad (1.1)$$

Second, the generation of this potential by a matter density ρ is governed by Poisson's differential equation:

$$\nabla^2\Phi = 4\pi G\rho, \quad (1.2)$$

where G is Newton's gravitational constant.

In General Relativity, we must replace these statements with laws that describe how the curvature of spacetime acts on matter to manifest as gravity, and how energy and momentum influence spacetime to create curvature.

The Minimal-Coupling Principle

The foundation for generalizing laws of physics from flat Minkowski spacetime to curved spacetime rests upon the Einstein Equivalence Principle (EEP). The EEP asserts that in sufficiently small regions of spacetime, the laws of physics reduce to those of Special Relativity, making it impossible to locally detect a gravitational field. This universality of gravity suggests it is not a conventional force propagating through spacetime, but rather a fundamental feature of the background manifold itself.

This philosophy yields a direct operational recipe for translating physical laws into a curved-spacetime context, known as the **minimal-coupling principle**. The procedure consists of three steps:

1. Take a physical law valid in inertial coordinates in flat spacetime.
2. Write the law in a manifestly coordinate-invariant (tensorial) form.
3. Assert that this tensorial law remains true in curved spacetime.

Mathematically, this "Comma-Goes-to-Semicolon Rule" amounts to replacing the Minkowski metric $\eta_{\mu\nu}$ with the general metric $g_{\mu\nu}$, and replacing partial derivatives ∂_μ with covariant derivatives ∇_μ .

Derivation of the Geodesic Equation

Let us apply the minimal-coupling principle to the motion of a freely-falling particle. In flat spacetime, a free particle moves in a straight line, which is expressed by the vanishing of the second derivative of its parameterized path $x^\mu(\lambda)$:

$$\frac{d^2x^\mu}{d\lambda^2} = 0. \quad (1.3)$$

While $dx^\mu/d\lambda$ transforms as a valid vector (the tangent vector to the curve), the second derivative $d^2x^\mu/d\lambda^2$ does not transform as a tensor in general curvilinear coordinates. To apply our recipe, we must first rewrite equation (1.3) in a tensorial form using the directional derivative along the path.

Using the chain rule, the ordinary derivative with respect to λ can be written as an operator acting on the coordinates:

$$\frac{d}{d\lambda} = \frac{dx^\nu}{d\lambda} \partial_\nu. \quad (1.4)$$

Applying this to the tangent vector $\frac{dx^\mu}{d\lambda}$, equation (1.3) becomes:

$$\frac{dx^\nu}{d\lambda} \partial_\nu \left(\frac{dx^\mu}{d\lambda} \right) = 0. \quad (1.5)$$

We now apply the minimal-coupling principle by elevating the partial derivative to a covariant derivative ($\partial_\nu \rightarrow \nabla_\nu$):

$$\frac{dx^\nu}{d\lambda} \nabla_\nu \left(\frac{dx^\mu}{d\lambda} \right) = 0. \quad (1.6)$$

This is the condition that the tangent vector is parallel transported along its own path. We expand the covariant derivative acting on the vector $V^\mu = dx^\mu/d\lambda$:

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma_{\nu\rho}^\mu V^\rho. \quad (1.7)$$

Substituting this expansion back into our directional derivative:

$$\begin{aligned} \frac{dx^\nu}{d\lambda} \left[\partial_\nu \left(\frac{dx^\mu}{d\lambda} \right) + \Gamma_{\nu\rho}^\mu \left(\frac{dx^\rho}{d\lambda} \right) \right] &= 0 \\ \frac{dx^\nu}{d\lambda} \partial_\nu \left(\frac{dx^\mu}{d\lambda} \right) + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda} &= 0. \end{aligned} \quad (1.8)$$

Recognizing that $\frac{dx^\nu}{d\lambda} \partial_\nu$ is simply the total derivative $\frac{d}{d\lambda}$ along the curve, we recover the **geodesic equation**:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0, \quad (1.9)$$

where we have relabeled the dummy indices ν and ρ to ρ and σ respectively to match standard conventions. Thus, free particles in curved spacetime move along geodesics.

Intuition: The extra term involving the Christoffel symbols $\Gamma_{\rho\sigma}^\mu$ corrects for the apparent acceleration caused strictly by the curvature of the coordinate system. Geometrically, this equation demands that the curve is as "straight" as possible within the curved manifold.

Energy-Momentum Conservation

A straightforward second example is the conservation of energy and momentum. In flat spacetime, this is expressed as the vanishing divergence of the energy-momentum tensor:

$$\partial_\mu T^{\mu\nu} = 0. \quad (1.10)$$

Applying the minimal-coupling recipe, the appropriate generalization to curved spacetime immediately becomes:

$$\nabla_\mu T^{\mu\nu} = 0. \quad (1.11)$$

The Newtonian Limit

To prove that this geometric framework legitimately describes gravity, we must show that it reduces to Newtonian gravity in the appropriate limit. The **Newtonian limit** is defined by three strict requirements:

1. **Slow motion:** The particles move at velocities much less than the speed of light ($v \ll c$).
2. **Static field:** The gravitational field is unchanging with time.
3. **Weak field:** The gravitational field is a small perturbation on a flat spacetime background.

1. Simplifying the Geodesic Equation

Let us analyze the geodesic equation under these assumptions, using proper time τ as the affine parameter. The slow motion condition mathematically implies that the spatial components of the four-velocity are negligible compared to the time component:

$$\left| \frac{dx^i}{d\tau} \right| \ll \left| \frac{dt}{d\tau} \right|. \quad (1.12)$$

In the summation term of the geodesic equation, $\Gamma_{\rho\sigma}^{\mu} \frac{dx^{\rho}}{d\tau} \frac{dx^{\sigma}}{d\tau}$, the dominant term by far will be the one where both $\rho = 0$ and $\sigma = 0$. We may therefore drop the spatial velocity terms, simplifying the equation to:

$$\frac{d^2 x^{\mu}}{d\tau^2} + \Gamma_{00}^{\mu} \left(\frac{dt}{d\tau} \right)^2 = 0. \quad (1.13)$$

2. Evaluating the Christoffel Symbol

We must calculate the relevant connection coefficient, Γ_{00}^{μ} . By definition:

$$\Gamma_{00}^{\mu} = \frac{1}{2} g^{\mu\lambda} (\partial_0 g_{\lambda 0} + \partial_0 g_{0\lambda} - \partial_{\lambda} g_{00}). \quad (1.14)$$

The static field assumption mandates that all time derivatives of the metric vanish ($\partial_0 g_{\alpha\beta} = 0$). This eliminates the first two terms in the parentheses, yielding:

$$\Gamma_{00}^{\mu} = -\frac{1}{2} g^{\mu\lambda} \partial_{\lambda} g_{00}. \quad (1.15)$$

3. The Weak Field Approximation

The weak field assumption allows us to write the metric as the flat Minkowski metric $\eta_{\mu\nu}$ plus a small perturbation $h_{\mu\nu}$:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (1.16)$$

We work in inertial coordinates with signature $(-+++)$, so $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$. To compute equation (1.15), we need the inverse metric $g^{\mu\nu}$. We postulate that the inverse metric takes the form $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$, where indices on the perturbation are raised using the flat background metric ($h^{\mu\nu} = \eta^{\mu\rho} \eta^{\nu\sigma} h_{\rho\sigma}$). Let us verify this explicit expansion to linear order in h :

$$\begin{aligned} g^{\mu\lambda} g_{\lambda\nu} &= (\eta^{\mu\lambda} - h^{\mu\lambda})(\eta_{\lambda\nu} + h_{\lambda\nu}) \\ &= \eta^{\mu\lambda} \eta_{\lambda\nu} + \eta^{\mu\lambda} h_{\lambda\nu} - h^{\mu\lambda} \eta_{\lambda\nu} - h^{\mu\lambda} h_{\lambda\nu} \\ &= \delta_{\nu}^{\mu} + \eta^{\mu\lambda} h_{\lambda\nu} - (\eta^{\mu\rho} \eta^{\lambda\sigma} h_{\rho\sigma}) \eta_{\lambda\nu} + \mathcal{O}(h^2) \\ &= \delta_{\nu}^{\mu} + \eta^{\mu\lambda} h_{\lambda\nu} - \eta^{\mu\rho} \delta_{\nu}^{\rho} h_{\rho\sigma} \\ &= \delta_{\nu}^{\mu} + \eta^{\mu\lambda} h_{\lambda\nu} - \eta^{\mu\rho} h_{\rho\nu}. \end{aligned} \quad (1.17)$$

The linear terms perfectly cancel (by relabeling dummy index $\rho \rightarrow \lambda$), verifying that $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$ is the correct inverse to first order in the perturbation.

We now substitute this into our Christoffel symbol. Since $\partial_{\lambda} g_{00} = \partial_{\lambda} (\eta_{00} + h_{00}) = \partial_{\lambda} h_{00}$ is already of order $\mathcal{O}(h)$, the term $-h^{\mu\lambda}$ from the inverse metric would generate a negligible $\mathcal{O}(h^2)$ term. Thus, we may simply replace the inverse metric with the flat metric:

$$\Gamma_{00}^{\mu} \approx -\frac{1}{2} \eta^{\mu\lambda} \partial_{\lambda} h_{00}. \quad (1.18)$$

Substituting this back into the slow-motion geodesic equation (1.13):

$$\frac{d^2 x^{\mu}}{d\tau^2} = \frac{1}{2} \eta^{\mu\lambda} \partial_{\lambda} h_{00} \left(\frac{dt}{d\tau} \right)^2. \quad (1.19)$$

4. Decomposing Time and Spatial Components

We examine the components of equation (1.19) separately. **Time Component** ($\mu = 0$):

$$\frac{d^2 t}{d\tau^2} = \frac{1}{2} \eta^{0\lambda} \partial_\lambda h_{00} \left(\frac{dt}{d\tau} \right)^2 = \frac{1}{2} \eta^{00} \partial_0 h_{00} \left(\frac{dt}{d\tau} \right)^2. \quad (1.20)$$

Because the field is static, $\partial_0 h_{00} = 0$, leading to:

$$\frac{d^2 t}{d\tau^2} = 0 \quad \implies \quad \frac{dt}{d\tau} = \text{constant}. \quad (1.21)$$

This confirms that in the Newtonian limit, coordinate time t ticks at a constant rate relative to proper time τ .

Spatial Components ($\mu = i$): The spacelike components of the Minkowski inverse metric are simply the 3×3 Euclidean identity matrix, $\eta^{ij} = \delta^{ij}$.

$$\frac{d^2 x^i}{d\tau^2} = \frac{1}{2} \eta^{ij} \partial_j h_{00} \left(\frac{dt}{d\tau} \right)^2 = \frac{1}{2} \delta^{ij} \partial_j h_{00} \left(\frac{dt}{d\tau} \right)^2 = \frac{1}{2} \partial_i h_{00} \left(\frac{dt}{d\tau} \right)^2. \quad (1.22)$$

We wish to write this equation purely in terms of coordinate time t , replacing the proper time derivatives. Using the chain rule rigorously:

$$\frac{d}{d\tau} = \frac{dt}{d\tau} \frac{d}{dt} \quad \implies \quad \frac{d^2 x^i}{d\tau^2} = \frac{d}{d\tau} \left(\frac{dt}{d\tau} \frac{dx^i}{dt} \right). \quad (1.23)$$

Applying the product rule gives:

$$\frac{d^2 x^i}{d\tau^2} = \left(\frac{d^2 t}{d\tau^2} \right) \frac{dx^i}{dt} + \frac{dt}{d\tau} \frac{d}{d\tau} \left(\frac{dx^i}{dt} \right). \quad (1.24)$$

We established that $d^2 t/d\tau^2 = 0$. Re-applying the chain rule to the second term:

$$\frac{d^2 x^i}{d\tau^2} = \left(\frac{dt}{d\tau} \right)^2 \frac{d^2 x^i}{dt^2}. \quad (1.25)$$

Substituting this back into our spatial equation allows the factors of $(dt/d\tau)^2$ to be canceled entirely:

$$\left(\frac{dt}{d\tau} \right)^2 \frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00} \left(\frac{dt}{d\tau} \right)^2 \quad \implies \quad \frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00}. \quad (1.26)$$

This is the equation of motion for a particle in a weak, static gravitational field.

5. Connection to the Newtonian Potential

We compare our geometric result with the classical Newtonian equation of motion from equation (1.1):

$$\frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00} \quad \text{vs} \quad \frac{d^2 x^i}{dt^2} = -\partial_i \Phi. \quad (1.27)$$

For these two formalisms to describe the exact same physical reality, we must identify the metric perturbation with the Newtonian potential:

$$\frac{1}{2} \partial_i h_{00} = -\partial_i \Phi \quad \implies \quad h_{00} = -2\Phi. \quad (1.28)$$

Since $g_{00} = \eta_{00} + h_{00}$ and we are using the signature where $\eta_{00} = -1$, the full time-time component of the metric is:

$$g_{00} = -(1 + 2\Phi). \quad (1.29)$$

This proves that the curvature of spacetime is sufficient to reproduce Newtonian gravity, provided the metric takes this specific form.

1.2 Einstein's Equation

Having established the geometrical framework and the response of test particles to curved spacetime, we turn to the second fundamental task of General Relativity: determining the field equations that govern how the metric $g_{\mu\nu}$ responds to the presence of energy and momentum.

Like Maxwell's equations in electrodynamics, the field equation of gravity cannot be rigorously derived from bedrock mathematical principles; it must be postulated and tested against experiment.

1.2.1 Motivating the Tensorial Equation

We seek a relativistic tensor equation that supersedes the classical Poisson equation for the Newtonian gravitational potential Φ :

$$\nabla^2 \Phi = 4\pi G \rho, \quad (1.30)$$

where $\nabla^2 = \delta^{ij} \partial_i \partial_j$ is the spatial Laplacian, ρ is the mass density, and G is Newton's gravitational constant.

To generalize this into a coordinate-invariant tensorial form, we analyze its components:

- **The Source (Right-Hand Side):** The mass density ρ generalizes cleanly to the energy-momentum tensor $T_{\mu\nu}$.
- **The Field (Left-Hand Side):** We established previously that the Newtonian potential Φ is embedded within the metric tensor perturbation, $h_{00} = -2\Phi$.

Therefore, we expect the relativistic generalization to relate a symmetric $(0, 2)$ tensor, constructed from second derivatives of the metric, to the energy-momentum tensor $T_{\mu\nu}$.

$$[\nabla^2 g]_{\mu\nu} \propto T_{\mu\nu}. \quad (1.31)$$

Our first impulse might be to apply the d'Alembertian operator $\square = \nabla^\mu \nabla_\mu$ to the metric $g_{\mu\nu}$.

Fortunately, differential geometry provides a unique, non-trivial tensor constructed from up to second derivatives of the metric: the Riemann curvature tensor $R^\rho{}_{\sigma\mu\nu}$.

The Riemann tensor is a rank-4 tensor, whereas $T_{\mu\nu}$ is rank-2. We must contract indices to match ranks.

$$R_{\mu\nu} = \kappa T_{\mu\nu}, \quad (1.32)$$

where κ is an undetermined coupling constant.

1.2.2 The Conservation of Energy Problem

Equation (1.32) contains a fatal flaw regarding conservation laws. In a closed system, local energy and momentum must be conserved, which is expressed geometrically by the vanishing covariant divergence of the energy-momentum tensor:

$$\nabla^\mu T_{\mu\nu} = 0. \quad (1.33)$$

If our proposed field equation $R_{\mu\nu} = \kappa T_{\mu\nu}$ were true, taking the covariant divergence of both sides would require:

$$\nabla^\mu R_{\mu\nu} = 0. \quad (1.34)$$

However, this is not true for a general Riemannian manifold.

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R, \quad (1.35)$$

where $R = g^{\mu\nu} R_{\mu\nu}$ is the Ricci scalar.

Let us explicitly show the contradiction. If $R_{\mu\nu} = \kappa T_{\mu\nu}$, then taking the trace of both sides yields:

$$g^{\mu\nu} R_{\mu\nu} = \kappa g^{\mu\nu} T_{\mu\nu} \implies R = \kappa T. \quad (1.36)$$

Substitute this into the Bianchi identity (1.35):

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu (\kappa T). \quad (1.37)$$

But we assumed $\nabla^\mu R_{\mu\nu} = \kappa \nabla^\mu T_{\mu\nu} = 0$. Therefore, we are forced to conclude:

$$\frac{1}{2} \kappa \nabla_\nu T = 0 \implies \nabla_\nu T = 0. \quad (1.38)$$

Since the covariant derivative of a scalar is simply its partial derivative ($\partial_\nu T = 0$), this implies that the trace of the energy-momentum tensor, T , must be a universal constant throughout all of spacetime.

1.2.3 The Einstein Tensor

To resolve this, we must replace the Ricci tensor on the left-hand side with a symmetric $(0, 2)$ tensor whose covariant divergence vanishes identically, irrespective of the specific metric.

We rearrange the contracted Bianchi identity (1.35) by moving all terms to one side:

$$\nabla^\mu R_{\mu\nu} - \frac{1}{2} \nabla_\nu R = 0. \quad (1.39)$$

We can rewrite $\nabla_\nu R$ using the metric tensor, since $\nabla^\mu g_{\mu\nu} = \delta_\nu^\mu$:

$$\nabla^\mu R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \nabla^\mu R = 0 \implies \nabla^\mu \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) = 0. \quad (1.40)$$

The quantity in the parentheses is the **Einstein Tensor**, $G_{\mu\nu}$:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \quad (1.41)$$

By mathematical necessity, the Einstein tensor is always divergence-free ($\nabla^\mu G_{\mu\nu} = 0$).

$$G_{\mu\nu} = \kappa T_{\mu\nu}. \quad (1.42)$$

This equation satisfies all our logical requirements: it equates a symmetric, divergence-free geometric tensor (containing up to second derivatives of the metric) to the symmetric, conserved energy-momentum tensor of matter.

1.2.4 The Trace-Reversed Form

To determine the coupling constant κ , it is highly advantageous to rewrite equation (1.42) by taking its trace. In a 4-dimensional spacetime, we contract both sides with $g^{\mu\nu}$:

$$\begin{aligned} g^{\mu\nu} G_{\mu\nu} &= \kappa g^{\mu\nu} T_{\mu\nu} \\ g^{\mu\nu} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) &= \kappa T \\ R - \frac{1}{2} R (g^{\mu\nu} g_{\mu\nu}) &= \kappa T. \end{aligned} \quad (1.43)$$

Since the trace of the metric in 4 dimensions is $g^{\mu\nu} g_{\mu\nu} = \delta_\mu^\mu = 4$, we have:

$$R - \frac{1}{2} R (4) = \kappa T \implies R - 2R = \kappa T \implies R = -\kappa T. \quad (1.44)$$

We substitute this expression for the Ricci scalar back into the original field equation $R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu}$:

$$R_{\mu\nu} - \frac{1}{2} (-\kappa T) g_{\mu\nu} = \kappa T_{\mu\nu}. \quad (1.45)$$

Rearranging to isolate the Ricci tensor yields the **trace-reversed form** of Einstein's Equation:

$$R_{\mu\nu} = \kappa \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right). \quad (1.46)$$

Equations (1.42) and (1.46) are mathematically identical; one is simply an algebraic rearrangement of the other.

1.2.5 The Newtonian Limit and the Coupling Constant

To find κ , we evaluate the trace-reversed equation (1.46) in the Newtonian limit, which requires a weak, static gravitational field, and slowly moving particles.

Consider the source of gravity to be a perfect fluid:

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}, \quad (1.47)$$

where ρ is the rest-frame energy density, p is the pressure, and U^μ is the fluid four-velocity.

$$T_{\mu\nu} = \rho U_\mu U_\nu. \quad (1.48)$$

We evaluate this in the rest frame of the massive fluid (e.g., a star), so the spatial components of the four-velocity vanish: $U^\mu = (U^0, 0, 0, 0)$.

To find U^0 , we enforce the normalization condition $g_{\mu\nu}U^\mu U^\nu = -1$.

$$g_{00}(U^0)^2 = -1 \implies (-1 + h_{00})(U^0)^2 = -1 \implies (U^0)^2 = (1 - h_{00})^{-1}. \quad (1.49)$$

Taylor expanding $(1 - h_{00})^{-1} \approx 1 + h_{00}$ to first order in the small perturbation h , we find:

$$U^0 \approx 1 + \frac{1}{2}h_{00}. \quad (1.50)$$

We must also find the covariant component $U_0 = g_{0\nu}U^\nu = g_{00}U^0$:

$$U_0 = (-1 + h_{00}) \left(1 + \frac{1}{2}h_{00} \right) \approx -1 - \frac{1}{2}h_{00} + h_{00} = -1 + \frac{1}{2}h_{00}. \quad (1.51)$$

However, since the energy density ρ is already considered small in this nearly-flat limit, retaining the metric perturbation $\mathcal{O}(h)$ inside the velocity terms will only produce negligible second-order $\mathcal{O}(\rho h)$ corrections.

We substitute these into the 00-component of the energy-momentum tensor:

$$T_{00} = \rho(U_0)^2 = \rho(-1)^2 = \rho. \quad (1.52)$$

All other components T_{0i} and T_{ij} vanish to this order. The trace $T = g^{\mu\nu}T_{\mu\nu}$ is dominated entirely by the time component:

$$T = g^{00}T_{00} \approx (-1)\rho = -\rho. \quad (1.53)$$

We now plug $T_{00} = \rho$ and $T = -\rho$ into the 00-component of our trace-reversed field equation (1.46):

$$\begin{aligned} R_{00} &= \kappa \left(T_{00} - \frac{1}{2}Tg_{00} \right) \\ R_{00} &\approx \kappa \left[\rho - \frac{1}{2}(-\rho)(-1) \right] \\ R_{00} &\approx \kappa \left(\rho - \frac{1}{2}\rho \right) = \frac{1}{2}\kappa\rho. \end{aligned} \quad (1.54)$$

Evaluating the Ricci Component R_{00}

To link equation (1.46) to the Newtonian potential, we must explicitly calculate R_{00} in terms of the metric perturbation $h_{\mu\nu}$.

$$R_{00} = R^\lambda_{0\lambda 0} = R^0_{000} + R^i_{0i0}. \quad (1.55)$$

Because the Riemann tensor is antisymmetric in its first two indices ($R^0_{000} = -R^0_{000}$), the term R^0_{000} vanishes identically.

$$R^i_{0j0} = \partial_j \Gamma^i_{00} - \partial_0 \Gamma^i_{j0} + \Gamma^i_{j\lambda} \Gamma^\lambda_{00} - \Gamma^i_{0\lambda} \Gamma^\lambda_{j0}. \quad (1.56)$$

We apply our Newtonian limit conditions: 1. **Static Field:** All time derivatives vanish ($\partial_0 \rightarrow 0$), so $\partial_0 \Gamma_{j0}^i = 0$. 2. **Weak Field:** The Christoffel symbols Γ are strictly linear in the metric perturbation $h_{\mu\nu}$. Therefore, the quadratic products $(\Gamma)^2$ are of order $\mathcal{O}(h^2)$ and can be neglected.

This drastically simplifies the Riemann component to just the first term:

$$R^i{}_{0j0} \approx \partial_j \Gamma_{00}^i. \quad (1.57)$$

Substituting the static, weak-field Christoffel symbol $\Gamma_{00}^i \approx -\frac{1}{2}\delta^{ij}\partial_j h_{00}$ into our expression for the Ricci tensor gives:

$$\begin{aligned} R_{00} &= R^i{}_{0i0} = \partial_i(\Gamma_{00}^i) \\ &= \partial_i \left(-\frac{1}{2}\delta^{ij}\partial_j h_{00} \right) \\ &= -\frac{1}{2}\delta^{ij}\partial_i\partial_j h_{00} = -\frac{1}{2}\nabla^2 h_{00}. \end{aligned} \quad (1.58)$$

Extracting Newton's Constant

We equate our two derived expressions for R_{00} :

$$-\frac{1}{2}\nabla^2 h_{00} = \frac{1}{2}\kappa\rho \quad \implies \quad \nabla^2 h_{00} = -\kappa\rho. \quad (1.59)$$

Recall our earlier identification linking the metric perturbation to the Newtonian gravitational potential: $h_{00} = -2\Phi$.

$$\nabla^2(-2\Phi) = -\kappa\rho \quad \implies \quad 2\nabla^2\Phi = \kappa\rho \quad \implies \quad \nabla^2\Phi = \frac{1}{2}\kappa\rho. \quad (1.60)$$

To perfectly match the classical Poisson equation, $\nabla^2\Phi = 4\pi G\rho$, the coupling constant must be set to $\kappa = 8\pi G$.

Substituting this value of κ back into our proposed equation (1.42) gives the final result—**Einstein's Field Equation**:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu}. \quad (1.61)$$

This profound equation tells us how the curvature of spacetime reacts to the presence of energy-momentum.

1.2.6 Vacuum Equations

We will often be interested in Einstein's equation in a vacuum, such as the region outside a star or planet. If $T_{\mu\nu} = 0$, the right-hand side of the trace-reversed equation (1.46) vanishes entirely.

$$R_{\mu\nu} = 0. \quad (1.62)$$

This is a considerably less formidable and highly useful form of the field equations for finding vacuum metrics.

Lagrangian Formulation in Curved Spacetime

1. The Physics: Why the Action Looks Like That

In classical mechanics, a particle takes a path that minimizes (or makes stationary) the action. In field theory, the “path” is the configuration of a field $\Phi(x)$ across all of spacetime.

- **The Flat Space Blueprint:** In flat space, the action is $S = \int \mathcal{L} d^n x$.

- **The Curved Space Reality:** In general relativity, the coordinate grid can stretch and warp. A coordinate volume $d^n x$ might represent a huge physical volume in one place and a tiny one in another.

To ensure our physics doesn't change just because our coordinate grid stretched, the action S must be a true, coordinate-independent number (a scalar).

1. We replace the flat-space Lagrangian with a true scalar $\hat{\mathcal{L}}$.
2. We replace regular partial derivatives ∂_μ with covariant derivatives ∇_μ because the field's rate of change must account for the curvature of the space it lives in.
3. We multiply by $\sqrt{-g}$ to convert the coordinate volume $d^n x$ into the invariant physical volume $\sqrt{-g} d^n x$.

2. The Math: Integration by Parts in Curved Space

To derive the equations of motion, we need to know how to move a covariant derivative from one term to another inside an integral.

Step A: The Product Rule Take two arbitrary objects, A and B , where one is a vector and the other a scalar (or they combine to form a vector). The covariant derivative obeys the standard Leibniz product rule:

$$\nabla_\mu(AB) = (\nabla_\mu A)B + A(\nabla_\mu B) \quad (1.63)$$

Step B: Rearranging We isolate the term we want to integrate by parts:

$$(\nabla_\mu A)B = \nabla_\mu(AB) - A(\nabla_\mu B) \quad (1.64)$$

Step C: Integrating and Stokes' Theorem Now, integrate all three terms over the physical spacetime volume:

$$\int (\nabla_\mu A)B\sqrt{-g} d^n x = \int \nabla_\mu(AB)\sqrt{-g} d^n x - \int A(\nabla_\mu B)\sqrt{-g} d^n x \quad (1.65)$$

The middle term is the integral of a total divergence. According to the generalized Stokes' theorem (the Divergence Theorem), the volume integral of a divergence turns into a surface integral over the boundary of that volume. Because we assume our field variations vanish at the boundaries of our universe (or at infinity), that boundary term is zero. We are left with the integration by parts formula:

$$\int (\nabla_\mu A)B\sqrt{-g} d^n x = - \int A(\nabla_\mu B)\sqrt{-g} d^n x \quad (1.66)$$

3. Deriving the Euler-Lagrange Equations

We vary the action $S = \int \hat{\mathcal{L}}\sqrt{-g} d^n x$ with respect to the field Φ by introducing a tiny perturbation: $\Phi \rightarrow \Phi + \delta\Phi$. We require the change in the action (δS) to be zero. Using the multivariable chain rule:

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi} \delta\Phi + \frac{\partial \hat{\mathcal{L}}}{\partial (\nabla_\mu \Phi)} \delta(\nabla_\mu \Phi) \right] \sqrt{-g} d^n x = 0 \quad (1.67)$$

Because the variation δ and the derivative ∇_μ commute, we write $\delta(\nabla_\mu \Phi) = \nabla_\mu(\delta\Phi)$.

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi} \delta\Phi + \frac{\partial \hat{\mathcal{L}}}{\partial (\nabla_\mu \Phi)} \nabla_\mu(\delta\Phi) \right] \sqrt{-g} d^n x = 0 \quad (1.68)$$

Applying our integration by parts formula to the second term (setting $A = \delta\Phi$ and $B = \frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\Phi)}$) gives:

$$\delta S = \int \left[\frac{\partial\hat{\mathcal{L}}}{\partial\Phi} \delta\Phi - \nabla_\mu \left(\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\Phi)} \right) \delta\Phi \right] \sqrt{-g} d^n x = 0 \quad (1.69)$$

Factoring out $\delta\Phi$:

$$\delta S = \int \left[\frac{\partial\hat{\mathcal{L}}}{\partial\Phi} - \nabla_\mu \left(\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\Phi)} \right) \right] \delta\Phi \sqrt{-g} d^n x = 0 \quad (1.70)$$

For this integral to vanish for *any* arbitrary variation $\delta\Phi$, the term inside the brackets must be identically zero everywhere. This yields the Euler-Lagrange equation.

4. Expanding the Scalar Field Example

The scalar Lagrangian is given by:

$$\hat{\mathcal{L}} = -\frac{1}{2} g^{\alpha\beta} (\nabla_\alpha \phi) (\nabla_\beta \phi) - V(\phi) \quad (1.71)$$

(Note: We use α and β as dummy indices to avoid confusion with the μ in the Euler-Lagrange equation.)

Step 1: Derivative with respect to ϕ The kinetic term doesn't contain ϕ directly, so we only differentiate the potential:

$$\frac{\partial\hat{\mathcal{L}}}{\partial\phi} = -\frac{dV}{d\phi} \quad (1.72)$$

Step 2: Derivative with respect to $\nabla_\mu\phi$ We apply the product rule to the kinetic term:

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2} g^{\alpha\beta} \left[\frac{\partial(\nabla_\alpha\phi)}{\partial(\nabla_\mu\phi)} (\nabla_\beta\phi) + (\nabla_\alpha\phi) \frac{\partial(\nabla_\beta\phi)}{\partial(\nabla_\mu\phi)} \right] \quad (1.73)$$

The derivative $\frac{\partial(\nabla_\alpha\phi)}{\partial(\nabla_\mu\phi)}$ is the Kronecker delta δ_α^μ :

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2} g^{\alpha\beta} \left[\delta_\alpha^\mu (\nabla_\beta\phi) + (\nabla_\alpha\phi) \delta_\beta^\mu \right] \quad (1.74)$$

The deltas force the indices to match μ :

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2} \left[g^{\mu\beta} \nabla_\beta\phi + g^{\alpha\mu} \nabla_\alpha\phi \right] \quad (1.75)$$

Because the metric is symmetric ($g^{\alpha\mu} = g^{\mu\alpha}$), these two terms are identical and cancel the 1/2:

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -g^{\mu\nu} \nabla_\nu\phi \quad (1.76)$$

Step 3: Plug into the Euler-Lagrange Equation Substitute our results from Step 1 and Step 2 into the master equation:

$$-\frac{dV}{d\phi} - \nabla_\mu (-g^{\mu\nu} \nabla_\nu\phi) = 0 \quad (1.77)$$

Because the metric is compatible with the covariant derivative ($\nabla_\mu g^{\mu\nu} = 0$), we can pull the metric outside the derivative:

$$-\frac{dV}{d\phi} + g^{\mu\nu} \nabla_\mu \nabla_\nu\phi = 0 \quad (1.78)$$

Recognizing $g^{\mu\nu} \nabla_\mu \nabla_\nu$ as the d'Alembertian operator $\square$, we arrive at the final equation of motion:

$$\square\phi - \frac{dV}{d\phi} = 0 \quad (1.79)$$

Summary Table

1.2.7 The Hilbert Action

We now turn to constructing an action for gravity itself.

Because the Equivalence Principle ensures that the metric can always be set to its canonical Minkowski form ($\eta_{\mu\nu}$) and its first derivatives can be set to zero ($\partial_\lambda g_{\mu\nu} = 0$) at any single point, any non-trivial scalar must involve at least *second* derivatives of the metric.

It can be rigorously proven that any non-trivial tensor made from products of the metric and its derivatives up to second order can be expressed entirely in terms of the metric and the Riemann tensor.

Recognizing this uniqueness, David Hilbert proposed that the Ricci scalar was the simplest and most natural choice for the gravitational Lagrangian.

$$S_H = \int \sqrt{-g} R d^n x. \quad (1.80)$$

1.2.8 Varying the Hilbert Action

To find the vacuum field equations, we must find the stationary points of the Hilbert action by applying the variational principle: $\delta S_H = 0$.

It is computationally more convenient to vary the action with respect to the *inverse* metric, $\delta g^{\mu\nu}$.

$$(\delta g^{\mu\lambda})g_{\lambda\nu} + g^{\mu\lambda}(\delta g_{\lambda\nu}) = 0. \quad (1.81)$$

Multiplying by $g^{\nu\sigma}$ isolates $\delta g_{\mu\nu}$:

$$\delta g_{\mu\nu} = -g_{\mu\rho}g_{\nu\sigma}\delta g^{\rho\sigma}. \quad (1.82)$$

Thus, finding stationary points with respect to $\delta g^{\mu\nu}$ is entirely equivalent to finding them with respect to $\delta g_{\mu\nu}$.

We expand the Ricci scalar as $R = g^{\mu\nu} R_{\mu\nu}$.

$$\begin{aligned} \delta S_H &= \delta \int d^n x \sqrt{-g} (g^{\mu\nu} R_{\mu\nu}) \\ &= \int d^n x [\sqrt{-g} g^{\mu\nu} (\delta R_{\mu\nu}) + \sqrt{-g} R_{\mu\nu} (\delta g^{\mu\nu}) + R (g^{\mu\nu} R_{\mu\nu}) (\delta \sqrt{-g})] \\ &\equiv (\delta S)_1 + (\delta S)_2 + (\delta S)_3. \end{aligned} \quad (1.83)$$

The middle term, $(\delta S)_2 = \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu}$, is already in the desired form (an expression multiplied by $\delta g^{\mu\nu}$).

Evaluating $(\delta S)_1$: Variation of the Ricci Tensor and the Boundary Term

The first term requires evaluating the variation of the Ricci tensor, $\delta R_{\mu\nu}$, which originates from the variation of the Riemann tensor. The Riemann tensor is defined entirely by the Levi-Civita connection (Christoffel symbols):

$$R^\rho{}_{\mu\lambda\nu} = \partial_\lambda \Gamma^\rho_{\nu\mu} - \partial_\nu \Gamma^\rho_{\lambda\mu} + \Gamma^\rho_{\lambda\sigma} \Gamma^\sigma_{\nu\mu} - \Gamma^\rho_{\nu\sigma} \Gamma^\sigma_{\lambda\mu}. \quad (1.84)$$

We evaluate the variation by treating the connection as the fundamental variable, shifting it by a small amount: $\Gamma^\rho_{\nu\mu} \rightarrow \Gamma^\rho_{\nu\mu} + \delta \Gamma^\rho_{\nu\mu}$. Crucially, while a single connection Γ is not a tensor, the *difference* between two connections, $\delta \Gamma^\rho_{\nu\mu}$, **is** a valid tensor. Because it is a tensor, we can take its covariant derivative.

The covariant derivative of this variation tensor is:

$$\nabla_\lambda (\delta \Gamma^\rho_{\nu\mu}) = \partial_\lambda (\delta \Gamma^\rho_{\nu\mu}) + \Gamma^\rho_{\lambda\sigma} \delta \Gamma^\sigma_{\nu\mu} - \Gamma^\sigma_{\lambda\nu} \delta \Gamma^\rho_{\sigma\mu} - \Gamma^\sigma_{\lambda\mu} \delta \Gamma^\rho_{\nu\sigma}. \quad (1.85)$$

By explicitly calculating the first-order variation of the Riemann tensor $\delta R^\rho{}_{\mu\lambda\nu}$ from its definition and comparing it to equation (1.98), the non-derivative terms perfectly cancel, yielding the **Palatini identity**:

$$\delta R^\rho{}_{\mu\lambda\nu} = \nabla_\lambda (\delta \Gamma^\rho_{\nu\mu}) - \nabla_\nu (\delta \Gamma^\rho_{\lambda\mu}). \quad (1.86)$$

Contracting the first and third indices ($\rho = \lambda$) gives the variation of the Ricci tensor:

$$\delta R_{\mu\nu} = \nabla_\rho(\delta\Gamma_{\nu\mu}^\rho) - \nabla_\nu(\delta\Gamma_{\rho\mu}^\rho). \quad (1.87)$$

We substitute this into the integral for $(\delta S)_1$:

$$(\delta S)_1 = \int d^n x \sqrt{-g} g^{\mu\nu} [\nabla_\rho(\delta\Gamma_{\nu\mu}^\rho) - \nabla_\nu(\delta\Gamma_{\rho\mu}^\rho)]. \quad (1.88)$$

Because the Levi-Civita connection is metric-compatible ($\nabla_\rho g^{\mu\nu} = 0$), we can pull the inverse metric inside the covariant derivatives. Relabeling dummy indices allows us to factor the expression into a single covariant divergence of a vector field V^σ :

$$(\delta S)_1 = \int d^n x \sqrt{-g} \nabla_\sigma [g^{\mu\nu}(\delta\Gamma_{\mu\nu}^\sigma) - g^{\mu\sigma}(\delta\Gamma_{\lambda\mu}^\lambda)] \equiv \int d^n x \sqrt{-g} \nabla_\sigma V^\sigma. \quad (1.89)$$

To rigorously prove that this term does not contribute to the equations of motion, we must express the vector V^σ purely in terms of the metric variation $\delta g_{\mu\nu}$. The variation of the Christoffel symbol in terms of the metric is:

$$\delta\Gamma_{\mu\nu}^\sigma = \frac{1}{2} g^{\sigma\rho} (\nabla_\mu \delta g_{\nu\rho} + \nabla_\nu \delta g_{\mu\rho} - \nabla_\rho \delta g_{\mu\nu}). \quad (1.90)$$

Substituting this into the two components of V^σ yields:

$$g^{\mu\nu} \delta\Gamma_{\mu\nu}^\sigma = \frac{1}{2} g^{\mu\nu} g^{\sigma\rho} (2\nabla_\mu \delta g_{\nu\rho} - \nabla_\rho \delta g_{\mu\nu}) = \nabla_\mu (\delta g^{\sigma\mu}) - \nabla^\sigma (\delta g_\mu^\mu), \quad (1.91)$$

$$g^{\mu\sigma} \delta\Gamma_{\lambda\mu}^\lambda = \frac{1}{2} g^{\mu\sigma} g^{\lambda\rho} (\nabla_\lambda \delta g_{\mu\rho} + \nabla_\mu \delta g_{\lambda\rho} - \nabla_\rho \delta g_{\lambda\mu}) = \frac{1}{2} \nabla^\sigma (\delta g_\lambda^\lambda) + \frac{1}{2} \nabla^\sigma (\delta g_\lambda^\lambda) - \frac{1}{2} \nabla^\sigma (\delta g_\lambda^\lambda) = \nabla^\sigma (\delta g_\mu^\mu). \quad (1.92)$$

Subtracting these results provides the explicit form of the boundary vector:

$$V^\sigma = \nabla_\mu (\delta g^{\sigma\mu}) - \nabla^\sigma (\delta g_\mu^\mu). \quad (1.93)$$

Thus, $(\delta S)_1$ is the spacetime integral of a pure covariant divergence. By the generalized Stokes's theorem, this volume integral converts perfectly into a surface integral over the boundary of the spacetime manifold $\partial\mathcal{M}$:

$$(\delta S)_1 = \int_{\mathcal{M}} d^n x \sqrt{-g} \nabla_\sigma V^\sigma = \int_{\partial\mathcal{M}} d^{n-1} y \sqrt{|h|} n_\sigma [\nabla_\mu (\delta g^{\sigma\mu}) - \nabla^\sigma (\delta g_\mu^\mu)], \quad (1.94)$$

where n_σ is the normal vector to the boundary and h is the determinant of the induced boundary metric.

In the standard formulation of the variational principle (Dirichlet boundary conditions), we demand that the metric variation and its first derivatives vanish on the boundary ($\delta g_{\mu\nu}|_{\partial\mathcal{M}} = 0$ and $\nabla_\rho \delta g_{\mu\nu}|_{\partial\mathcal{M}} = 0$). Under these strict boundary conditions, the entire surface integral evaluates to exactly zero. Therefore, the first term contributes nothing to the dynamical field equations:

$$(\delta S)_1 = 0. \quad (1.95)$$

1.3 The Palatini Formulation and Identity

1.3.1 Proof the Palatini Identity

To vary the action with respect to the connection, we must understand how the Riemann and Ricci tensors change under a small variation $\Gamma_{\mu\nu}^\rho \rightarrow \Gamma_{\mu\nu}^\rho + \delta\Gamma_{\mu\nu}^\rho$.

By definition, the Riemann tensor constructed from an arbitrary connection is:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda. \quad (1.96)$$

Before varying, we must establish a crucial tensorial property. Under a general coordinate transformation $x \rightarrow x'$, a connection transforms as:

$$\Gamma_{\mu'\nu'}^{\rho'} = \frac{\partial x^{\rho'}}{\partial x^{\rho}} \frac{\partial x^{\mu}}{\partial x^{\mu'}} \frac{\partial x^{\nu}}{\partial x^{\nu'}} \Gamma_{\mu\nu}^{\rho} + \frac{\partial x^{\rho'}}{\partial x^{\lambda}} \frac{\partial^2 x^{\lambda}}{\partial x^{\mu'} \partial x^{\nu'}}. \quad (1.97)$$

Because of the second term involving the second derivative of the coordinates, the connection itself is *not* a tensor. However, if we take the difference of two valid connections defined on the same manifold, $\delta\Gamma_{\mu\nu}^{\rho} = \hat{\Gamma}_{\mu\nu}^{\rho} - \Gamma_{\mu\nu}^{\rho}$, the non-tensorial second-derivative terms perfectly subtract out. Thus, the variation $\delta\Gamma_{\mu\nu}^{\rho}$ is a **true tensor of rank (1,2)**.

Because $\delta\Gamma_{\mu\nu}^{\rho}$ is a tensor, we can meaningfully take its covariant derivative with respect to the background connection Γ . The expansion of this covariant derivative is:

$$\nabla_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) = \partial_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) + \Gamma_{\mu\lambda}^{\rho} \delta\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\mu\nu}^{\lambda} \delta\Gamma_{\lambda\sigma}^{\rho} - \Gamma_{\mu\sigma}^{\lambda} \delta\Gamma_{\nu\lambda}^{\rho}. \quad (1.98)$$

We now apply the variation directly to the definition of the Riemann tensor in equation (1.96). Dropping terms of order $\mathcal{O}(\delta\Gamma^2)$, we obtain the first-order variation:

$$\delta R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) - \partial_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}) + (\delta\Gamma_{\mu\lambda}^{\rho})\Gamma_{\nu\sigma}^{\lambda} + \Gamma_{\mu\lambda}^{\rho}(\delta\Gamma_{\nu\sigma}^{\lambda}) - (\delta\Gamma_{\nu\lambda}^{\rho})\Gamma_{\mu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho}(\delta\Gamma_{\mu\sigma}^{\lambda}). \quad (1.99)$$

We aim to express this variation not in terms of partial derivatives, but in terms of the covariant derivatives we defined in equation (1.98). Let us explicitly calculate the difference of two covariant derivatives of our variation tensor:

$$\begin{aligned} \nabla_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) - \nabla_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}) &= \left[\partial_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) + \Gamma_{\mu\lambda}^{\rho} \delta\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\mu\nu}^{\lambda} \delta\Gamma_{\lambda\sigma}^{\rho} - \Gamma_{\mu\sigma}^{\lambda} \delta\Gamma_{\nu\lambda}^{\rho} \right] \\ &\quad - \left[\partial_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}) + \Gamma_{\nu\lambda}^{\rho} \delta\Gamma_{\mu\sigma}^{\lambda} - \Gamma_{\nu\mu}^{\lambda} \delta\Gamma_{\lambda\sigma}^{\rho} - \Gamma_{\nu\sigma}^{\lambda} \delta\Gamma_{\mu\lambda}^{\rho} \right]. \end{aligned} \quad (1.100)$$

We distribute the negative sign and group the terms:

$$\begin{aligned} \nabla_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) - \nabla_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}) &= \partial_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) - \partial_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}) \\ &\quad + \Gamma_{\mu\lambda}^{\rho} \delta\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \delta\Gamma_{\mu\sigma}^{\lambda} \\ &\quad - \Gamma_{\mu\nu}^{\lambda} \delta\Gamma_{\lambda\sigma}^{\rho} + \Gamma_{\nu\mu}^{\lambda} \delta\Gamma_{\lambda\sigma}^{\rho} \\ &\quad - \Gamma_{\mu\sigma}^{\lambda} \delta\Gamma_{\nu\lambda}^{\rho} + \Gamma_{\nu\sigma}^{\lambda} \delta\Gamma_{\mu\lambda}^{\rho}. \end{aligned} \quad (1.101)$$

Because we assumed the connection is torsion-free ($\Gamma_{\mu\nu}^{\lambda} = \Gamma_{\nu\mu}^{\lambda}$), the two terms in the third line cancel each other exactly. Rearranging the remaining terms yields:

$$\nabla_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) - \nabla_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}) = \partial_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) - \partial_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}) + \Gamma_{\mu\lambda}^{\rho} \delta\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \delta\Gamma_{\mu\sigma}^{\lambda} + \Gamma_{\nu\sigma}^{\lambda} \delta\Gamma_{\mu\lambda}^{\rho} - \Gamma_{\mu\sigma}^{\lambda} \delta\Gamma_{\nu\lambda}^{\rho}. \quad (1.102)$$

Comparing this result term-by-term with equation (1.99), we find an exact match. This establishes the **Palatini identity** for the variation of the Riemann tensor:

$$\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_{\mu}(\delta\Gamma_{\nu\sigma}^{\rho}) - \nabla_{\nu}(\delta\Gamma_{\mu\sigma}^{\rho}). \quad (1.103)$$

To find the variation of the Ricci tensor, we contract the first and third indices ($\rho = \mu$). Relabeling the remaining indices to conventional forms (letting $\sigma \rightarrow \mu$ and $\nu \rightarrow \nu$), we obtain the variation of the Ricci tensor:

$$\delta R_{\mu\nu} = \nabla_{\lambda}(\delta\Gamma_{\mu\nu}^{\lambda}) - \nabla_{\nu}(\delta\Gamma_{\mu\lambda}^{\lambda}). \quad (1.104)$$

This beautifully compact equation demonstrates that the variation of the Ricci tensor is simply the difference of two covariant divergences.

1.3.2 Variation of the Action and Integration by Parts

We now return to the Palatini action in equation (??). The total variation of the action δS must vanish for arbitrary variations of both $g^{\mu\nu}$ and $\Gamma_{\mu\nu}^\rho$.

Varying with respect to $g^{\mu\nu}$ trivially recovers the standard algebraic relation $R_{\mu\nu}(\Gamma) - \frac{1}{2}R(\Gamma)g_{\mu\nu} = 0$, exactly as in the Einstein-Hilbert derivation.

The profound physics emerges when we vary the action with respect to the connection Γ :

$$\delta_\Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu}. \quad (1.105)$$

We substitute the Palatini identity (1.104) into the integral:

$$\delta_\Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} \left[\nabla_\lambda (\delta \Gamma_{\mu\nu}^\lambda) - \nabla_\nu (\delta \Gamma_{\mu\lambda}^\lambda) \right]. \quad (1.106)$$

To isolate $\delta \Gamma$, we must integrate by parts. However, because Γ and g are assumed to be fully independent, we **cannot** assume that $\nabla_\lambda g^{\mu\nu} = 0$. We must treat the term $\sqrt{-g} g^{\mu\nu}$ as a general tensor density and rigorously apply the Leibniz rule for covariant derivatives.

Let us define the tensor density $\mathfrak{g}^{\mu\nu} = \sqrt{-g} g^{\mu\nu}$. The product rule dictates:

$$\nabla_\lambda (\mathfrak{g}^{\mu\nu} \delta \Gamma_{\mu\nu}^\lambda) = (\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma_{\mu\nu}^\lambda + \mathfrak{g}^{\mu\nu} \nabla_\lambda (\delta \Gamma_{\mu\nu}^\lambda). \quad (1.107)$$

Rearranging this to solve for the term appearing in our action:

$$\mathfrak{g}^{\mu\nu} \nabla_\lambda (\delta \Gamma_{\mu\nu}^\lambda) = \nabla_\lambda (\mathfrak{g}^{\mu\nu} \delta \Gamma_{\mu\nu}^\lambda) - (\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma_{\mu\nu}^\lambda. \quad (1.108)$$

A similar expansion applies to the second term in equation (1.106):

$$\mathfrak{g}^{\mu\nu} \nabla_\nu (\delta \Gamma_{\mu\lambda}^\lambda) = \nabla_\nu (\mathfrak{g}^{\mu\nu} \delta \Gamma_{\mu\lambda}^\lambda) - (\nabla_\nu \mathfrak{g}^{\mu\nu}) \delta \Gamma_{\mu\lambda}^\lambda. \quad (1.109)$$

Substituting these into the variation of the action yields:

$$\delta_\Gamma S = \int d^4x \left[\nabla_\lambda (\mathfrak{g}^{\mu\nu} \delta \Gamma_{\mu\nu}^\lambda) - \nabla_\nu (\mathfrak{g}^{\mu\nu} \delta \Gamma_{\mu\lambda}^\lambda) - (\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma_{\mu\nu}^\lambda + (\nabla_\nu \mathfrak{g}^{\mu\nu}) \delta \Gamma_{\mu\lambda}^\lambda \right]. \quad (1.110)$$

The first two terms are total covariant divergences of vector densities. For a torsion-free connection, the covariant divergence of any vector density $\mathcal{V}^\lambda$ reduces exactly to an ordinary partial divergence: $\nabla_\lambda \mathcal{V}^\lambda = \partial_\lambda \mathcal{V}^\lambda$. By Stokes's theorem, the volume integral of a partial divergence transforms into a boundary integral at infinity. Assuming the variations $\delta \Gamma$ vanish at the boundary of spacetime, these total derivative terms evaluate strictly to zero.

We are left entirely with the terms where the covariant derivative acts on the metric density:

$$\delta_\Gamma S = \int d^4x \left[-(\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma_{\mu\nu}^\lambda + (\nabla_\nu \mathfrak{g}^{\mu\nu}) \delta \Gamma_{\mu\lambda}^\lambda \right] = 0. \quad (1.111)$$

To factor out $\delta \Gamma_{\mu\nu}^\lambda$, we must align the dummy indices in the second term. We swap $\lambda \leftrightarrow \nu$, yielding $(\nabla_\lambda \mathfrak{g}^{\mu\lambda}) \delta \Gamma_{\mu\nu}^\nu$. Since $\delta \Gamma_{\mu\nu}^\nu = \delta_\lambda^\nu \delta \Gamma_{\mu\nu}^\lambda$, we can write:

$$\delta_\Gamma S = \int d^4x \left[-\nabla_\lambda \mathfrak{g}^{\mu\nu} + \delta_\lambda^\nu \nabla_\sigma \mathfrak{g}^{\mu\sigma} \right] \delta \Gamma_{\mu\nu}^\lambda = 0. \quad (1.112)$$

Because the connection is symmetric, the variation $\delta \Gamma_{\mu\nu}^\lambda$ is symmetric in its lower indices (μ, ν) . Therefore, for the integral to vanish for arbitrary variations, the *symmetric part* of the bracketed expression must vanish identically:

$$-\nabla_\lambda \mathfrak{g}^{\mu\nu} + \frac{1}{2} \delta_\lambda^\nu \nabla_\sigma \mathfrak{g}^{\mu\sigma} + \frac{1}{2} \delta_\lambda^\mu \nabla_\sigma \mathfrak{g}^{\nu\sigma} = 0. \quad (1.113)$$

1.3.3 Recovering Metric Compatibility

Equation (1.113) acts as the dynamical equation of motion for the connection. We must solve it to find the explicit form of $\nabla_\lambda \mathbf{g}^{\mu\nu}$.

We take the trace of equation (1.113) by contracting the indices λ and ν ($\lambda = \nu$). Remembering that in 4-dimensional spacetime, the trace of the Kronecker delta is $\delta^\nu_\nu = 4$:

$$\begin{aligned} -\nabla_\nu \mathbf{g}^{\mu\nu} + \frac{1}{2}(4)\nabla_\sigma \mathbf{g}^{\mu\sigma} + \frac{1}{2}\delta^\mu_\nu \nabla_\sigma \mathbf{g}^{\nu\sigma} &= 0 \\ -\nabla_\sigma \mathbf{g}^{\mu\sigma} + 2\nabla_\sigma \mathbf{g}^{\mu\sigma} + \frac{1}{2}\nabla_\sigma \mathbf{g}^{\mu\sigma} &= 0 \\ \frac{3}{2}\nabla_\sigma \mathbf{g}^{\mu\sigma} &= 0. \end{aligned} \tag{1.114}$$

This proves that the vector density must be divergence-free: $\nabla_\sigma \mathbf{g}^{\mu\sigma} = 0$.

We substitute this crucial result back into our primary constraint equation (1.113). Because $\nabla_\sigma \mathbf{g}^{\mu\sigma} = 0$, the second and third terms in the equation vanish entirely, leaving the strict requirement:

$$\nabla_\lambda \mathbf{g}^{\mu\nu} = 0 \quad \implies \quad \nabla_\lambda (\sqrt{-g} g^{\mu\nu}) = 0. \tag{1.115}$$

To prove that this mandates full metric compatibility, we apply the product rule to the density:

$$\sqrt{-g} \nabla_\lambda g^{\mu\nu} + g^{\mu\nu} \nabla_\lambda \sqrt{-g} = 0. \tag{1.116}$$

We can find the covariant derivative of the determinant factor by investigating the determinant of the density $\mathbf{g}^{\mu\nu}$ itself. The determinant of the matrix $\sqrt{-g} g^{\mu\nu}$ in 4 dimensions is:

$$\det(\mathbf{g}^{\mu\nu}) = (\sqrt{-g})^4 \det(g^{\mu\nu}) = g^2 (g^{-1}) = g. \tag{1.117}$$

Since $\nabla_\lambda \mathbf{g}^{\mu\nu} = 0$, the covariant derivative of its matrix determinant must also vanish, meaning $\nabla_\lambda g = 0$. Since g is a constant under covariant differentiation, $\sqrt{-g}$ must also be constant:

$$\nabla_\lambda \sqrt{-g} = 0. \tag{1.118}$$

Substituting this back into equation (1.116) annihilates the second term, yielding the final, monumental result:

$$\sqrt{-g} \nabla_\lambda g^{\mu\nu} = 0 \quad \implies \quad \nabla_\lambda g^{\mu\nu} = 0. \tag{1.119}$$

Evaluating $(\delta S)_3$: Variation of the Determinant

To evaluate the third term, we must find the variation of the metric determinant, δg .

$$\ln(\det M) = \text{Tr}(\ln M). \tag{1.120}$$

Varying this identity with respect to M yields Jacobi's formula:

$$\frac{1}{\det M} \delta(\det M) = \text{Tr}(M^{-1} \delta M). \tag{1.121}$$

Applying this to the metric tensor matrix ($M = g_{\mu\nu}$, $\det M = g$, $M^{-1} = g^{\mu\nu}$):

$$\frac{1}{g} \delta g = \text{Tr}(g^{\mu\nu} \delta g_{\mu\nu}) = g^{\mu\nu} \delta g_{\mu\nu}. \tag{1.122}$$

We convert the variation to be with respect to the inverse metric using equation (1.82):

$$\delta g = g(g^{\mu\nu} \delta g_{\mu\nu}) = g(g^{\mu\nu} [-g_{\mu\rho} g_{\nu\sigma} \delta g^{\rho\sigma}]) = -g(\delta^\nu_\rho g_{\nu\sigma} \delta g^{\rho\sigma}) = -g(g_{\rho\sigma} \delta g^{\rho\sigma}). \tag{1.123}$$

Relabeling indices gives $\delta g = -g(g_{\mu\nu}\delta g^{\mu\nu})$.

$$\begin{aligned}
\delta\sqrt{-g} &= \frac{1}{2\sqrt{-g}}(-\delta g) \\
&= \frac{1}{2\sqrt{-g}}(g(g_{\mu\nu}\delta g^{\mu\nu})) \\
&= -\frac{1}{2}\left(\frac{-g}{\sqrt{-g}}\right)g_{\mu\nu}\delta g^{\mu\nu} \\
&= -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}.
\end{aligned} \tag{1.124}$$

We substitute this result back into the third term of the action variation:

$$(\delta S)_3 = \int d^n x R(\delta\sqrt{-g}) = \int d^n x \sqrt{-g} \left(-\frac{1}{2}Rg_{\mu\nu}\right) \delta g^{\mu\nu}. \tag{1.125}$$

1.3.4 Deriving the Vacuum Equations

We now assemble the full variation δS_H by combining the non-vanishing components $(\delta S)_2$ and $(\delta S)_3$:

$$\begin{aligned}
\delta S_H &= \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \int d^n x \sqrt{-g} \left(-\frac{1}{2}Rg_{\mu\nu}\right) \delta g^{\mu\nu} \\
\delta S_H &= \int d^n x \sqrt{-g} \left[R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right] \delta g^{\mu\nu}.
\end{aligned} \tag{1.126}$$

The general definition of a functional derivative for an action S with respect to a set of fields Φ^i is:

$$\delta S = \int \sum_i \left(\frac{\delta S}{\delta \Phi^i} \delta \Phi^i\right) d^n x. \tag{1.127}$$

Comparing this definition to our expression for δS_H , where our field is $g^{\mu\nu}$, we identify the functional derivative of the Hilbert action:

$$\frac{1}{\sqrt{-g}} \frac{\delta S_H}{\delta g^{\mu\nu}} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}. \tag{1.128}$$

The principle of least action demands that the action is stationary ($\delta S_H = 0$) for arbitrary variations $\delta g^{\mu\nu}$.

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 0. \tag{1.129}$$

The power of the Lagrangian approach is now evident.

1.3.5 Including Matter: The Energy-Momentum Tensor

To obtain the complete, non-vacuum Einstein equation, we must augment the gravitational Hilbert action with an action describing the matter fields, S_M .

$$S = \frac{1}{16\pi G} S_H + S_M, \tag{1.130}$$

where we have explicitly included the normalization constant $1/(16\pi G)$ on the gravitational term to ensure we recover the correct Newtonian limit.

We apply the variational principle to the total action with respect to the inverse metric:

$$\delta S = \frac{1}{16\pi G} \delta S_H + \delta S_M = 0. \tag{1.131}$$

Dividing by the volume measure $\sqrt{-g}$ and utilizing our previous result for δS_H :

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}} = \frac{1}{16\pi G} \left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right) + \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}} = 0. \tag{1.132}$$

Rearranging this equation isolates the geometric terms on the left:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = -16\pi G \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}. \quad (1.133)$$

By comparing this to the known form of Einstein's equation ($G_{\mu\nu} = 8\pi GT_{\mu\nu}$), we are led to a profound, fundamental definition of the energy-momentum tensor $T_{\mu\nu}$ for any matter field:

$$T_{\mu\nu} \equiv \frac{-2}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}. \quad (1.134)$$

Verification via the Scalar Field

To verify that equation (1.134) is a sensible physical definition, we apply it to the scalar field action defined in equation (??).

$$\begin{aligned} \delta S_\phi &= \delta \int d^n x \left[\sqrt{-g} \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \right] \\ &= \int d^n x \left[\sqrt{-g} \delta \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) + \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \delta \sqrt{-g} \right]. \end{aligned} \quad (1.135)$$

In the first term, the only explicit metric dependence is the inverse metric $g^{\rho\sigma}$, so its variation is $-\frac{1}{2}\delta g^{\mu\nu}(\nabla_\mu \phi \nabla_\nu \phi)$.

$$\delta S_\phi = \int d^n x \sqrt{-g} \delta g^{\mu\nu} \left[-\frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \right]. \quad (1.136)$$

We extract the functional derivative $\delta S_\phi / \delta g^{\mu\nu}$ and apply our new definition (1.134):

$$\begin{aligned} T_{\mu\nu}^{(\phi)} &= \frac{-2}{\sqrt{-g}} \frac{\delta S_\phi}{\delta g^{\mu\nu}} \\ &= -2 \left[-\frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi + \frac{1}{4} g_{\mu\nu} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi + \frac{1}{2} g_{\mu\nu} V(\phi) \right] \\ T_{\mu\nu}^{(\phi)} &= \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - g_{\mu\nu} V(\phi). \end{aligned} \quad (1.137)$$

In the flat spacetime limit ($g_{\mu\nu} \rightarrow \eta_{\mu\nu}$, $\nabla \rightarrow \partial$), this precisely recovers the standard energy-momentum tensor for a scalar field derived in classical field theory.

The Hilbert vs. Canonical Tensor

In flat spacetime field theory, Noether's theorem defines a "canonical" energy-momentum tensor $S^{\mu\nu}$ associated with symmetry under spacetime translations:

$$S^{\mu\nu} = \frac{\delta \mathcal{L}}{\delta(\partial_\mu \Phi^i)} \partial^\nu \Phi^i - \eta^{\mu\nu} \mathcal{L}. \quad (1.138)$$

While $S^{\mu\nu}$ is conserved ($\partial_\mu S^{\mu\nu} = 0$), it suffers from significant drawbacks: it is not always symmetric ($S^{\mu\nu} \neq S^{\nu\mu}$), it is not guaranteed to be gauge-invariant, and it cannot always be generalized to curved spacetime.

In stark contrast, the Hilbert definition (1.134) derived from the metric variation is vastly superior for General Relativity.

1.3.6 Properties of Einstein's Equation

We conclude by outlining the mathematical nature of Einstein's Equation.

While it appears to be a system of ten independent equations (since $G_{\mu\nu}$ and $T_{\mu\nu}$ are symmetric 4×4 matrices), the contracted Bianchi identity ($\nabla^\mu G_{\mu\nu} = 0$) imposes four differential constraints on the system.

This under-determination of four equations is not a mathematical flaw, but a physical necessity.

Finally, the equations are highly non-linear.

1.4 Properties of Einstein's Equation

Einstein's equation functions as a system of second-order partial differential equations determining the metric tensor field $g_{\mu\nu}$. To fully utilize this framework, we must rigorously analyze its mathematical properties, its degrees of freedom, its inherent non-linearity, and how it governs the physical motion of matter.

1.4.1 Degrees of Freedom and Diffeomorphism Invariance

Because both the Einstein tensor $G_{\mu\nu}$ and the energy-momentum tensor $T_{\mu\nu}$ are symmetric $(0, 2)$ tensors, Einstein's equation appears to constitute ten independent differential equations. This seemingly provides a perfectly determined system for the ten independent component functions of the metric tensor.

However, the geometry of the manifold imposes constraints. The contracted Bianchi identity, $\nabla^\mu G_{\mu\nu} = 0$, enforces four differential constraints on the Ricci tensor and scalar. Consequently, there are only six truly independent dynamical equations governing the metric.

Mathematical and Physical Interpretation: This under-determination is not a flaw, but a mathematically necessary reflection of the theory's coordinate independence (diffeomorphism invariance). If a specific metric $g_{\mu\nu}(x)$ solves the field equations in a coordinate system x^μ , the tensor transformation law guarantees it will also be a solution in any smoothly transformed coordinate system $x^{\mu'}(x)$. The freedom to arbitrarily choose these four transformation functions $x^{\mu'}(x^\mu)$ perfectly accounts for the four "missing" constraining equations. General Relativity only determines the six physical, coordinate-independent degrees of freedom of the gravitational field.

1.4.2 The Non-Linearity of Gravity

A defining characteristic of General Relativity is its strict non-linearity. In Newtonian gravity, the gravitational potential of a composite system is the linear superposition of the individual potentials of its constituents. In General Relativity, the principle of superposition fails entirely.

Mathematically, the field equations are non-linear because the Ricci scalar and tensor are constructed from the Riemann tensor, which contains products of the Christoffel symbols $(\Gamma_{\mu\nu}^\rho \Gamma_{\alpha\beta}^\sigma)$. These symbols inherently contain the inverse metric multiplied by first derivatives of the metric.

Physically, this non-linearity arises because the gravitational field couples to itself. The Equivalence Principle dictates that all forms of energy and momentum gravitate. Because the gravitational field itself contains energy, it must act as its own source.

Intuition Example (Perturbative Gravity): In quantum electrodynamics (a linear theory), the electromagnetic interaction between two electrons is mediated by the exchange of a virtual photon. Because photons carry no electric charge, they do not couple to one another. In contrast, the hypothetical quantum force carrier of gravity, the graviton, couples to energy-momentum. Therefore, gravitons interact with other gravitons. A massive body does not just pull on a test particle; the gravitational field of the body pulls on its own gravitational field, compounding the spacetime curvature. This back-reaction is what causes deviations from Newtonian orbits, such as the anomalous perihelion precession of Mercury.

1.4.3 Raychaudhuri's Equation and the Attractiveness of Gravity

To understand exactly how the energy-momentum distribution dictates the physical motion of matter, we consider the geometric evolution of a small, initially stationary ball of free test particles. The particles follow timelike geodesics with four-velocities U^μ . The fractional rate of change of the volume of this ball is quantified by the expansion scalar, $\theta = \nabla_\mu U^\mu$.

The purely geometric evolution of this expansion is governed by Raychaudhuri's equation:

$$\frac{d\theta}{d\tau} = 2\omega^2 - 2\sigma^2 - \frac{1}{3}\theta^2 - R_{\mu\nu}U^\mu U^\nu, \quad (1.139)$$

where τ is proper time, ω^2 measures the rotation (vorticity), and σ^2 measures the shear (volume-preserving distortion).

Let us evaluate this system at an initial moment where all particles in the ball are mutually at rest. The initial expansion, rotation, and shear strictly vanish: $\theta = \omega = \sigma = 0$. We construct a locally inertial coordinate system $x^{\hat{\mu}}$ at the center of the ball, aligning the time axis with the particles' rest frame so that $U^{\hat{\mu}} = (1, 0, 0, 0)$. Raychaudhuri's equation simplifies immensely to:

$$\frac{d\theta}{d\tau} = -R_{\hat{0}\hat{0}}. \quad (1.140)$$

We now introduce the physical source via the trace-reversed Einstein equation:

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right). \quad (1.141)$$

In our locally inertial rest frame, the stress-energy tensor is decomposed into the energy density $\rho = T_{\hat{0}\hat{0}}$ and the principal spatial pressures $p_k = T_{\hat{k}\hat{k}}$. The trace is:

$$T = \eta^{\hat{\alpha}\hat{\beta}} T_{\hat{\alpha}\hat{\beta}} = -T_{\hat{0}\hat{0}} + T_{\hat{1}\hat{1}} + T_{\hat{2}\hat{2}} + T_{\hat{3}\hat{3}} = -\rho + p_x + p_y + p_z. \quad (1.142)$$

Substituting this trace into the 00-component of Einstein's equation (noting $\eta_{\hat{0}\hat{0}} = -1$):

$$\begin{aligned} R_{\hat{0}\hat{0}} &= 8\pi G \left[\rho - \frac{1}{2}(-\rho + p_x + p_y + p_z)(-1) \right] \\ R_{\hat{0}\hat{0}} &= 8\pi G \left[\rho - \frac{1}{2}\rho + \frac{1}{2}(p_x + p_y + p_z) \right] \\ R_{\hat{0}\hat{0}} &= 4\pi G(\rho + p_x + p_y + p_z). \end{aligned} \quad (1.143)$$

Equating this to the kinematic expansion yields the physical equation of motion for the volume:

$$\frac{d\theta}{d\tau} = -4\pi G(\rho + p_x + p_y + p_z). \quad (1.144)$$

Geometric and Physical Interpretation: Equation (1.144) proves that gravity is universally attractive ($d\theta/d\tau < 0$, causing the volume to shrink) as long as the sum $\rho + \sum p_i$ is positive. Remarkably, pressure acts as a direct source of gravitational attraction. This is a severe departure from Newtonian gravity, where Poisson's equation ($\nabla^2 \Phi = 4\pi G \rho$) depends exclusively on the mass density. While local pressure gradients exert physical forces that push matter apart, the uniform presence of pressure dynamically increases the strength of the attractive gravitational field.

1.4.4 Derivation of the Weyl Tensor Propagation Equation

The textbook concisely presents the divergence of the Weyl tensor and its relation to the energy-momentum tensor. However, deriving this propagation equation requires meticulously combining the second Bianchi identity with the full expansion of the Weyl tensor. We perform this explicit derivation here.

1. The Contracted Bianchi Identity

We begin with the second Bianchi identity for the Riemann curvature tensor:

$$\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\mu R_{\rho\sigma\nu\lambda} + \nabla_\nu R_{\rho\sigma\lambda\mu} = 0. \quad (1.145)$$

We wish to find the divergence of the Riemann tensor, $\nabla^\rho R_{\rho\sigma\mu\nu}$. We contract equation (1.145) with the inverse metric $g^{\rho\lambda}$ (setting $\lambda = \alpha, \rho = \alpha$ and summing):

$$\nabla^\rho R_{\rho\sigma\mu\nu} + \nabla_\mu R^\rho_{\sigma\nu\rho} + \nabla_\nu R^\rho_{\sigma\rho\mu} = 0. \quad (1.146)$$

Using the antisymmetry of the Riemann tensor ($R^\rho_{\sigma\nu\rho} = -R^\rho_{\sigma\rho\nu} = -R_{\sigma\nu}$), we evaluate the two traced terms:

$$\nabla_\mu R^\rho_{\sigma\nu\rho} = -\nabla_\mu R_{\sigma\nu} \quad (1.147)$$

$$\nabla_\nu R^\rho_{\sigma\rho\mu} = \nabla_\nu R_{\sigma\mu}. \quad (1.148)$$

Substituting these back yields the contracted Bianchi identity for the Riemann tensor:

$$\nabla^\rho R_{\rho\sigma\mu\nu} - \nabla_\mu R_{\sigma\nu} + \nabla_\nu R_{\sigma\mu} = 0 \implies \nabla^\rho R_{\rho\sigma\mu\nu} = \nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}. \quad (1.149)$$

Using anti-symmetrization bracket notation $A_{[\mu\nu]} = \frac{1}{2}(A_{\mu\nu} - A_{\nu\mu})$, this is compactly written as:

$$\nabla^\rho R_{\rho\sigma\mu\nu} = 2\nabla_{[\mu} R_{\nu]\sigma}. \quad (1.150)$$

2. The Divergence of the Weyl Tensor

The full Weyl tensor $C_{\rho\sigma\mu\nu}$ in a 4-dimensional spacetime is defined as the trace-free part of the Riemann tensor:

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{1}{2}(g_{\rho\mu}R_{\sigma\nu} - g_{\rho\nu}R_{\sigma\mu} - g_{\sigma\mu}R_{\rho\nu} + g_{\sigma\nu}R_{\rho\mu}) + \frac{1}{6}R(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}). \quad (1.151)$$

We take the covariant divergence ∇^ρ of this entire expression. Because the Levi-Civita connection is metric-compatible ($\nabla^\rho g_{\alpha\beta} = 0$), the metric tensors pass freely through the derivative:

$$\nabla^\rho C_{\rho\sigma\mu\nu} = \nabla^\rho R_{\rho\sigma\mu\nu} - \frac{1}{2}(\delta^\rho_\mu \nabla^\rho R_{\sigma\nu} - \delta^\rho_\nu \nabla^\rho R_{\sigma\mu} - g_{\sigma\mu} \nabla^\rho R_{\rho\nu} + g_{\sigma\nu} \nabla^\rho R_{\rho\mu}) + \frac{1}{6}(\delta^\rho_\mu g_{\sigma\nu} \nabla^\rho R - \delta^\rho_\nu g_{\sigma\mu} \nabla^\rho R). \quad (1.152)$$

We evaluate these contractions term by term:

- The first term is simply the contracted Bianchi identity (1.149): $\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}$.
- In the second group, the Kronecker deltas map $\rho \rightarrow \mu$ and $\rho \rightarrow \nu$: $\delta^\rho_\mu \nabla^\rho R_{\sigma\nu} = \nabla_\mu R_{\sigma\nu}$.
- Also in the second group, we apply the twice-contracted Bianchi identity ($\nabla^\rho R_{\rho\alpha} = \frac{1}{2}\nabla_\alpha R$) to the terms $g_{\sigma\mu} \nabla^\rho R_{\rho\nu} = \frac{1}{2}g_{\sigma\mu} \nabla_\nu R$.
- In the third group, the deltas again map ρ to the lower indices: $\delta^\rho_\mu g_{\sigma\nu} \nabla^\rho R = g_{\sigma\nu} \nabla_\mu R$.

Substituting all these evaluated terms back into the divergence gives:

$$\nabla^\rho C_{\rho\sigma\mu\nu} = (\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) - \frac{1}{2}\left(\nabla_\mu R_{\sigma\nu} - \nabla_\nu R_{\sigma\mu} - \frac{1}{2}g_{\sigma\mu} \nabla_\nu R + \frac{1}{2}g_{\sigma\nu} \nabla_\mu R\right) + \frac{1}{6}(g_{\sigma\nu} \nabla_\mu R - g_{\sigma\mu} \nabla_\nu R). \quad (1.153)$$

Since the Ricci tensor is symmetric ($R_{\mu\nu} = R_{\nu\mu}$), we can combine the Ricci derivative terms:

$$\begin{aligned} \nabla^\rho C_{\rho\sigma\mu\nu} &= \left(1 - \frac{1}{2}\right)(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) + \left(\frac{1}{4} - \frac{1}{6}\right)g_{\sigma\mu} \nabla_\nu R - \left(\frac{1}{4} - \frac{1}{6}\right)g_{\sigma\nu} \nabla_\mu R \\ &= \frac{1}{2}(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) + \frac{1}{12}(g_{\sigma\mu} \nabla_\nu R - g_{\sigma\nu} \nabla_\mu R). \end{aligned} \quad (1.154)$$

We rewrite this result using the anti-symmetrization brackets:

$$\frac{1}{2}(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) = \nabla_{[\mu} R_{\nu]\sigma} \quad (1.155)$$

$$\frac{1}{12}(g_{\sigma\mu} \nabla_\nu R - g_{\sigma\nu} \nabla_\mu R) = \frac{1}{6}\left[\frac{1}{2}(g_{\sigma\mu} \nabla_\nu R - g_{\sigma\nu} \nabla_\mu R)\right] = \frac{1}{6}g_{\sigma[\mu} \nabla_{\nu]} R. \quad (1.156)$$

(Note in (1.156) that swapping μ and ν absorbs the negative sign, giving $+g_{\sigma[\mu} \nabla_{\nu]} R$). Summing these yields the geometric identity for the propagation of the Weyl tensor:

$$\nabla^\rho C_{\rho\sigma\mu\nu} = \nabla_{[\mu} R_{\nu]\sigma} + \frac{1}{6}g_{\sigma[\mu} \nabla_{\nu]} R. \quad (1.157)$$

3. Introducing the Matter Source

We now couple this purely geometric identity to the matter distribution using the trace-reversed Einstein field equation:

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right). \quad (1.158)$$

Taking the trace of this equation gives the Ricci scalar: $R = -8\pi G T$.

We substitute these into the two terms on the right-hand side of equation (1.157). For the first term, we take the covariant derivative and anti-symmetrize:

$$\begin{aligned} \nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma} &= 8\pi G \left[\nabla_\mu \left(T_{\nu\sigma} - \frac{1}{2} T g_{\nu\sigma} \right) - \nabla_\nu \left(T_{\mu\sigma} - \frac{1}{2} T g_{\mu\sigma} \right) \right] \\ &= 8\pi G \left[(\nabla_\mu T_{\nu\sigma} - \nabla_\nu T_{\mu\sigma}) - \frac{1}{2} (g_{\nu\sigma} \nabla_\mu T - g_{\mu\sigma} \nabla_\nu T) \right]. \end{aligned} \quad (1.159)$$

Converting this to bracket notation:

$$2\nabla_{[\mu} R_{\nu]\sigma} = 8\pi G (2\nabla_{[\mu} T_{\nu]\sigma} - g_{\sigma[\nu} \nabla_{\mu]} T). \quad (1.160)$$

Because $g_{\sigma[\nu} \nabla_{\mu]} T = -g_{\sigma[\mu} \nabla_{\nu]} T$, we divide by 2 to obtain:

$$\nabla_{[\mu} R_{\nu]\sigma} = 8\pi G \left(\nabla_{[\mu} T_{\nu]\sigma} + \frac{1}{2} g_{\sigma[\mu} \nabla_{\nu]} T \right). \quad (1.161)$$

For the second term in (1.157), we simply substitute $R = -8\pi G T$:

$$\frac{1}{6} g_{\sigma[\mu} \nabla_{\nu]} R = \frac{1}{6} g_{\sigma[\mu} \nabla_{\nu]} (-8\pi G T) = 8\pi G \left(-\frac{1}{6} g_{\sigma[\mu} \nabla_{\nu]} T \right). \quad (1.162)$$

Finally, we sum equations (1.161) and (1.162) to assemble the full Weyl propagation equation:

$$\begin{aligned} \nabla^\rho C_{\rho\sigma\mu\nu} &= 8\pi G \left[\nabla_{[\mu} T_{\nu]\sigma} + \left(\frac{1}{2} - \frac{1}{6} \right) g_{\sigma[\mu} \nabla_{\nu]} T \right] \\ \nabla^\rho C_{\rho\sigma\mu\nu} &= 8\pi G \left(\nabla_{[\mu} T_{\nu]\sigma} + \frac{1}{3} g_{\sigma[\mu} \nabla_{\nu]} T \right). \end{aligned} \quad (1.163)$$

This explicitly demonstrates how gradients in the local energy, momentum, and pressure precisely dictate the radiation of tidal curvature across empty spacetime.

1.4.5 Quantum Gravity and the Planck Scale

A significant limitation of General Relativity is that it is a classical field theory. Standard quantization procedures applied to the metric yield a non-renormalizable theory; when considering higher-order quantum loops, ultraviolet divergences appear that cannot be absorbed by a finite number of parameters.

The breakdown of classical relativity is strictly isolated to extreme physical regimes determined by the **Planck Scale**, constructed from G , c , and $\hbar$:

$$\text{Planck Mass: } m_P = \left(\frac{\hbar c}{G} \right)^{1/2} \approx 1.22 \times 10^{19} \text{ GeV} \quad (1.164)$$

$$\text{Planck Length: } l_P = \left(\frac{\hbar G}{c^3} \right)^{1/2} \approx 1.62 \times 10^{-33} \text{ cm} \quad (1.165)$$

$$\text{Planck Time: } t_P = \left(\frac{\hbar G}{c^5} \right)^{1/2} \approx 5.39 \times 10^{-44} \text{ sec} \quad (1.166)$$

$$\text{Planck Energy: } E_P = \left(\frac{\hbar c^5}{G} \right)^{1/2} \approx 1.22 \times 10^{19} \text{ GeV} \quad (1.167)$$

At scales approaching l_P or E_P , the smooth manifold structure of spacetime is expected to be fundamentally altered by quantum fluctuations, requiring a complete theory of quantum gravity, such as String Theory.

To rigorously demonstrate that these specific combinations yield the correct dimensions, we can perform a dimensional analysis. Let the fundamental dimensions be denoted as Mass $[M]$, Length $[L]$, and Time $[T]$. The dimensions of the fundamental constants are:

$$[c] = LT^{-1} \quad (1.168)$$

$$[\hbar] = ML^2T^{-1} \quad (1.169)$$

$$[G] = M^{-1}L^3T^{-2} \quad (1.170)$$

Substituting these base dimensions into the equations for the Planck units proves their consistency:

$$[m_P] = \left[\frac{\hbar c}{G} \right]^{1/2} = \left(\frac{(ML^2T^{-1})(LT^{-1})}{M^{-1}L^3T^{-2}} \right)^{1/2} = \left(\frac{ML^3T^{-2}}{M^{-1}L^3T^{-2}} \right)^{1/2} = (M^2)^{1/2} = M \quad (1.171)$$

$$[l_P] = \left[\frac{\hbar G}{c^3} \right]^{1/2} = \left(\frac{(ML^2T^{-1})(M^{-1}L^3T^{-2})}{(LT^{-1})^3} \right)^{1/2} = \left(\frac{L^5T^{-3}}{L^3T^{-3}} \right)^{1/2} = (L^2)^{1/2} = L \quad (1.172)$$

$$[t_P] = \left[\frac{\hbar G}{c^5} \right]^{1/2} = \left(\frac{(ML^2T^{-1})(M^{-1}L^3T^{-2})}{(LT^{-1})^5} \right)^{1/2} = \left(\frac{L^5T^{-3}}{L^5T^{-5}} \right)^{1/2} = (T^2)^{1/2} = T \quad (1.173)$$

$$[E_P] = \left[\frac{\hbar c^5}{G} \right]^{1/2} = \left(\frac{(ML^2T^{-1})(L^5T^{-5})}{M^{-1}L^3T^{-2}} \right)^{1/2} = \left(\frac{ML^7T^{-6}}{M^{-1}L^3T^{-2}} \right)^{1/2} = (M^2L^4T^{-4})^{1/2} = ML^2T^{-2} \quad (1.174)$$

This matches the standard macroscopic dimensions for Energy ($[E] = [\text{Force} \times \text{Distance}] = ML^2T^{-2}$).

1.5 The Cosmological Constant

A fundamental and distinguishing characteristic of General Relativity is that the gravitational field couples to the entire energy-momentum tensor. In classical Newtonian mechanics and standard quantum field theory, the absolute zero-point of energy is physically arbitrary; only relative changes in energy between states dictate the dynamical equations of motion. Consequently, one can freely add a constant potential V_0 to the system without altering the physics. In gravitation, however, the absolute magnitude of the energy fundamentally matters. The total, un-normalized energy density acts as a direct source of spacetime curvature.

1.5.1 Vacuum Energy and Lorentz Invariance

This behavior inherently opens up the physical possibility of vacuum energy: an intrinsic, non-vanishing energy density characteristic of entirely empty space. If the vacuum is to be truly empty, it must not possess any preferred frame of reference or distinguished spatial direction. Therefore, if a non-zero energy density exists, its associated energy-momentum tensor must remain strictly Lorentz invariant in any locally inertial coordinate system.

In a locally inertial frame, the only $(0, 2)$ tensor that is invariant under the full Lorentz group is the Minkowski metric $\eta_{\hat{\mu}\hat{\nu}}$. Thus, the vacuum stress-energy tensor must be proportional to the metric:

$$T_{\hat{\mu}\hat{\nu}}^{(vac)} = -\rho_{vac}\eta_{\hat{\mu}\hat{\nu}}, \quad (1.175)$$

where ρ_{vac} is a constant scalar representing the vacuum energy density. By the principle of general covariance, we can elevate this locally inertial expression to a general coordinate-invariant tensor equation valid in any reference frame:

$$T_{\mu\nu}^{(vac)} = -\rho_{vac}g_{\mu\nu}. \quad (1.176)$$

Furthermore, the energy density ρ_{vac} must be a universal constant throughout spacetime, because any non-zero gradient ($\nabla_\lambda \rho_{vac} \neq 0$) would establish a preferred direction, breaking the assumed Lorentz invariance of the vacuum.

1.5.2 Equation of State for the Vacuum

We can deduce the physical properties of this vacuum energy by comparing equation (1.176) to the standard stress-energy tensor for a perfect fluid:

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}. \quad (1.177)$$

For the fluid tensor to strictly match the vacuum tensor $-\rho_{vac}g_{\mu\nu}$, the term proportional to the four-velocities $U_\mu U_\nu$ must identically vanish. This enforces the condition $\rho_{vac} + p_{vac} = 0$, which immediately yields the vacuum equation of state:

$$p_{vac} = -\rho_{vac}. \quad (1.178)$$

This establishes that a positive vacuum energy density necessitates a strictly isotropic negative pressure of equal magnitude.

1.5.3 Modifying Einstein's Equation

We can decompose the total energy-momentum of the universe into a standard matter/radiation piece $T_{\mu\nu}^{(M)}$ and the vacuum piece $T_{\mu\nu}^{(vac)}$. Substituting this decomposition into the standard field equation gives:

$$\begin{aligned} R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} &= 8\pi G \left(T_{\mu\nu}^{(M)} + T_{\mu\nu}^{(vac)} \right) \\ R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} &= 8\pi G \left(T_{\mu\nu}^{(M)} - \rho_{vac}g_{\mu\nu} \right). \end{aligned} \quad (1.179)$$

Rearranging the terms to place all geometric components on the left-hand side yields:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + 8\pi G\rho_{vac}g_{\mu\nu} = 8\pi GT_{\mu\nu}^{(M)}. \quad (1.180)$$

Historically, Einstein introduced a new term, the **cosmological constant** Λ , to construct a static cosmological model. The modified field equation takes the form:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi GT_{\mu\nu}. \quad (1.181)$$

By directly comparing equation (1.180) with equation (1.181), it is evident that the cosmological constant is mathematically and physically equivalent to a uniform vacuum energy density:

$$\rho_{vac} = \frac{\Lambda}{8\pi G}. \quad (1.182)$$

In modern theoretical physics, the terms "cosmological constant" and "vacuum energy" are utilized interchangeably.

1.5.4 Lagrangian Derivation

This modification can be derived rigorously from the principle of least action. The standard Hilbert Lagrangian density for gravity is $\hat{\mathcal{L}}_H = R$, which is the simplest scalar constructable from the metric. However, an even simpler scalar is a fundamental constant. We can augment the gravitational action by including a constant term -2Λ :

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} (R - 2\Lambda) + \hat{\mathcal{L}}_M \right]. \quad (1.183)$$

To verify this produces the correct field equation, we compute the variation of the new term with respect to the inverse metric $\delta g^{\mu\nu}$. Recalling the variation of the determinant $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$:

$$\begin{aligned} \delta S_\Lambda &= \delta \int d^4x \sqrt{-g} \left(\frac{-2\Lambda}{16\pi G} \right) \\ &= \int d^4x \left(\frac{-2\Lambda}{16\pi G} \right) \left(-\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu} \right) \\ &= \frac{1}{16\pi G} \int d^4x \sqrt{-g} (\Lambda g_{\mu\nu}) \delta g^{\mu\nu}. \end{aligned} \quad (1.184)$$

When combined with the variation of the Ricci scalar (which generates the Einstein tensor $G_{\mu\nu}$), this variation rigorously yields the $+\Lambda g_{\mu\nu}$ term on the left-hand side of Einstein's equation.

1.5.5 The Cosmological Constant Problem

While vacuum energy is easily accommodated within the mathematics of General Relativity, calculating its expected magnitude is highly problematic. In quantum field theory, the vacuum state possesses zero-point energy due to quantum fluctuations. Consider a simple harmonic oscillator with classical potential $V(x) = \frac{1}{2}\omega^2 x^2$. The Heisenberg uncertainty principle forbids the particle from resting exactly at the minimum with zero momentum, shifting the ground state energy to $E_0 = \frac{1}{2}\hbar\omega$.

A free quantum field can be decomposed via Fourier transform into an infinite collection of harmonic oscillators in momentum space. The frequency of each mode is $\omega = \sqrt{m^2 + k^2}$, where k is the wave vector magnitude. The total vacuum energy density is the integral over all momentum modes:

$$\rho_{vac} = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2} \hbar \sqrt{m^2 + k^2}. \quad (1.185)$$

This integral diverges to infinity. If we impose an ultraviolet momentum cutoff k_{max} (representing the scale at which our field theory breaks down), dimensional analysis indicates the vacuum energy density scales as:

$$\rho_{vac} \sim \hbar k_{max}^4. \quad (1.186)$$

If we assume classical field theory is valid up to the reduced Planck scale $\bar{m}_P \sim 10^{18}$ GeV, we calculate an immense vacuum energy density:

$$\rho_{vac}^{(theory)} \sim (10^{18} \text{ GeV})^4 \sim 10^{112} \text{ erg/cm}^3. \quad (1.187)$$

Conversely, cosmological observations of the universe's large-scale structure and expansion constrain the physical vacuum energy to be extremely small:

$$|\rho_{\Lambda}^{(obs)}| \leq (10^{-12} \text{ GeV})^4 \sim 10^{-8} \text{ erg/cm}^3. \quad (1.188)$$

The ratio between the theoretical expectation in equation (1.187) and the observational bound in equation (1.188) reveals a discrepancy of 120 orders of magnitude. Because no known symmetry naturally cancels this massive vacuum energy to such a precise, near-zero value, this massive theoretical failure is known as the "cosmological constant problem".

1.6 Energy Conditions

Einstein's field equation, $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, is fundamentally a system of differential equations determining the metric $g_{\mu\nu}$ for a given matter distribution $T_{\mu\nu}$. However, because the Einstein tensor satisfies the contracted Bianchi identity identically ($\nabla^\mu G_{\mu\nu} = 0$), one can technically postulate any arbitrary smooth metric $g_{\mu\nu}$, compute its corresponding Einstein tensor, and simply define that tensor to be $8\pi G T_{\mu\nu}$. Because of the Bianchi identity, the resulting energy-momentum tensor is guaranteed to be conserved.

This implies that any arbitrary spacetime geometry—including unphysical ones such as traversable wormholes or warp drives—is a mathematical solution to Einstein's equation for *some* hypothetical matter distribution. To filter out these unphysical spacetimes, we must impose coordinate-invariant restrictions on the energy-momentum tensor. These restrictions are known as **energy conditions**. They limit the arbitrariness of $T_{\mu\nu}$ by contracting it with physically meaningful vectors (timelike or null).

To develop physical intuition, we evaluate each energy condition for the case of a perfect fluid:

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + p g_{\mu\nu}, \quad (1.189)$$

where U^μ is the fluid four-velocity, ρ is the rest-frame energy density, and p is the isotropic pressure.

1.6.1 The Weak Energy Condition (WEC)

The Weak Energy Condition mandates that the local energy density measured by any physical observer must be non-negative. Since an observer's worldline is defined by a timelike tangent vector t^μ , the WEC states:

$$T_{\mu\nu}t^\mu t^\nu \geq 0 \quad \text{for all timelike vectors } t^\mu. \quad (1.190)$$

To translate this into conditions on ρ and p , we evaluate this contraction in a locally inertial reference frame where the fluid is at rest, meaning $U^\mu = (1, 0, 0, 0)$. In this frame, the metric is $g_{\mu\nu} = \eta_{\mu\nu}$, and the stress-energy tensor is diagonal: $T_{\mu\nu} = \text{diag}(\rho, p, p, p)$.

Let $t^\mu = (t^0, t^i)$ be an arbitrary timelike vector, which satisfies $t_\mu t^\mu < 0$. In our local frame:

$$-(t^0)^2 + (t^i)^2 < 0 \implies (t^i)^2 < (t^0)^2, \quad (1.191)$$

where $(t^i)^2 \equiv \delta_{ij}t^i t^j$. We can define a squared velocity parameter $v^2 = (t^i)^2 / (t^0)^2$, where $0 \leq v^2 < 1$.

The WEC contraction is:

$$\begin{aligned} T_{\mu\nu}t^\mu t^\nu &= T_{00}(t^0)^2 + T_{ij}t^i t^j \\ &= \rho(t^0)^2 + p(t^i)^2 \\ &= (t^0)^2 \left[\rho + p \frac{(t^i)^2}{(t^0)^2} \right] \\ &= (t^0)^2 (\rho + p v^2). \end{aligned} \quad (1.192)$$

For this expression to remain non-negative for *all* valid timelike observers (all $v^2 \in [0, 1)$), we must check the boundaries of the interval:

1. At $v = 0$ (an observer comoving with the fluid), we require $\rho \geq 0$.
2. In the limit $v \rightarrow 1$ (an observer moving at nearly the speed of light), we require $\rho + p \geq 0$.

Thus, the WEC dictates two physical constraints:

$$\rho \geq 0 \quad \text{and} \quad \rho + p \geq 0. \quad (1.193)$$

1.6.2 The Null Energy Condition (NEC)

The Null Energy Condition is a weaker version of the WEC, applied only to massless particles (photons). It states:

$$T_{\mu\nu}l^\mu l^\nu \geq 0 \quad \text{for all null vectors } l^\mu. \quad (1.194)$$

Using the same perfect fluid setup, a null vector satisfies $l_\mu l^\mu = 0$, meaning $(l^0)^2 = (l^i)^2$. The contraction becomes:

$$\begin{aligned} T_{\mu\nu}l^\mu l^\nu &= \rho(l^0)^2 + p(l^i)^2 \\ &= \rho(l^0)^2 + p(l^0)^2 \\ &= (l^0)^2 (\rho + p). \end{aligned} \quad (1.195)$$

Since $(l^0)^2 > 0$, the NEC imposes a single condition:

$$\rho + p \geq 0. \quad (1.196)$$

Notably, the NEC allows for a negative energy density ($\rho < 0$), provided there is a sufficiently large positive pressure to compensate.

1.6.3 The Dominant Energy Condition (DEC)

The Dominant Energy Condition incorporates the WEC, but adds the requirement that the flow of energy-momentum cannot exceed the speed of light. Physically, the four-momentum density measured by an observer t^μ is $P^\mu = -T^\mu_\nu t^\nu$. The DEC requires that P^μ is a causal (non-spacelike) vector pointing in the same time direction as t^μ :

$$T_{\mu\nu}t^\mu t^\nu \geq 0 \quad \text{and} \quad P_\mu P^\mu \leq 0 \quad \text{for all timelike } t^\mu. \quad (1.197)$$

Let us evaluate the causal constraint $P_\mu P^\mu \leq 0$ for a perfect fluid in its rest frame. The mixed tensor T^μ_ν is:

$$T^0_0 = g^{00}T_{00} = (-1)(\rho) = -\rho, \quad T^i_j = g^{ik}T_{kj} = \delta^{ik}(p\delta_{kj}) = p\delta^i_j. \quad (1.198)$$

The components of the momentum density P^μ are:

$$P^0 = -T^0_\nu t^\nu = -T^0_0 t^0 = -(-\rho)t^0 = \rho t^0 \quad (1.199)$$

$$P^i = -T^i_\nu t^\nu = -T^i_j t^j = -pt^i. \quad (1.200)$$

We calculate the invariant norm:

$$\begin{aligned} P_\mu P^\mu &= -(P^0)^2 + (P^i)^2 \\ &= -(\rho t^0)^2 + (-pt^i)^2 \\ &= -\rho^2(t^0)^2 + p^2(t^i)^2. \end{aligned} \quad (1.201)$$

For P^μ to be non-spacelike, we require $P_\mu P^\mu \leq 0$:

$$p^2(t^i)^2 \leq \rho^2(t^0)^2 \implies \frac{p^2}{\rho^2} \leq \frac{(t^0)^2}{(t^i)^2}. \quad (1.202)$$

Since t^μ is timelike, the ratio $(t^0)^2/(t^i)^2$ is strictly greater than 1. For the inequality to hold for *all* timelike vectors (even as $(t^i)^2 \rightarrow (t^0)^2$), we must satisfy the strictest limit:

$$\frac{p^2}{\rho^2} \leq 1 \implies p^2 \leq \rho^2. \quad (1.203)$$

Combining this with the WEC requirement ($\rho \geq 0$), the DEC is mathematically equivalent to:

$$\rho \geq |p|. \quad (1.204)$$

This ensures the energy density is non-negative and strictly bounds the magnitude of the pressure.

1.6.4 The Null Dominant Energy Condition (NDEC)

The Null Dominant Energy Condition enforces the DEC, but exclusively for null vectors l^μ :

$$T_{\mu\nu}l^\mu l^\nu \geq 0 \quad \text{and} \quad P_\mu P^\mu \leq 0 \quad \text{where } P^\mu = -T^\mu_\nu l^\nu. \quad (1.205)$$

Re-evaluating the invariant norm for a null vector ($(l^0)^2 = (l^i)^2$):

$$\begin{aligned} P_\mu P^\mu &= -\rho^2(l^0)^2 + p^2(l^i)^2 \\ &= (l^0)^2(p^2 - \rho^2). \end{aligned} \quad (1.206)$$

For $P_\mu P^\mu \leq 0$, we require $p^2 \leq \rho^2$, which yields $|\rho| \geq |p|$. Coupled with the NEC ($\rho + p \geq 0$), the NDEC permits negative energy densities ($\rho < 0$) under one highly constrained circumstance: if ρ is negative, the condition $|\rho| \geq |p|$ implies $-\rho \geq p$. However, the NEC requires $p \geq -\rho$. The only way both can be true is if $p = -\rho$.

Thus, the NDEC perfectly excludes all sources forbidden by the DEC, with the singular exception of a negative cosmological constant (negative vacuum energy).

1.6.5 The Strong Energy Condition (SEC)

The Strong Energy Condition differs structurally from the others by involving the trace of the energy-momentum tensor, $T = T^\lambda{}_\lambda$. It dictates:

$$T_{\mu\nu}t^\mu t^\nu \geq \frac{1}{2}Tt^\sigma t_\sigma \quad \text{for all timelike vectors } t^\mu. \quad (1.207)$$

For a perfect fluid, the trace is $T = -\rho + 3p$. The invariant length of the timelike vector is $t^\sigma t_\sigma = -(t^0)^2 + (t^i)^2 = -(t^0)^2(1 - v^2)$. Substituting these into the SEC inequality:

$$\begin{aligned} (t^0)^2(\rho + pv^2) &\geq \frac{1}{2}(-\rho + 3p) [-(t^0)^2(1 - v^2)] \\ \rho + pv^2 &\geq \frac{1}{2}(\rho - 3p)(1 - v^2). \end{aligned} \quad (1.208)$$

Evaluating this at the velocity limits:

1. At $v = 0$: $\rho \geq \frac{1}{2}(\rho - 3p) \implies 2\rho \geq \rho - 3p \implies \rho + 3p \geq 0$.
2. As $v \rightarrow 1$: $\rho + p \geq 0$.

The Strong Energy Condition does *not* imply the WEC, because it does not strictly require $\rho \geq 0$. It does, however, forbid large negative pressures.

The SEC and Gravitational Attraction

The mathematical form of the SEC has a profound geometric origin rooted in Raychaudhuri's equation. Recall that the purely geometric expansion of a family of geodesics is driven by the contraction of the Ricci tensor with the four-velocity: $d\theta/d\tau \approx -R_{\mu\nu}U^\mu U^\nu$.

Einstein's equation is $R_{\mu\nu} = 8\pi G(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu})$. Contracting this with arbitrary timelike vectors t^μ yields:

$$\begin{aligned} R_{\mu\nu}t^\mu t^\nu &= 8\pi G \left(T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tg_{\mu\nu}t^\mu t^\nu \right) \\ &= 8\pi G \left(T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tt_\sigma t^\sigma \right). \end{aligned} \quad (1.209)$$

For gravity to be universally attractive ($d\theta/d\tau \leq 0$), the quantity $R_{\mu\nu}t^\mu t^\nu$ must be non-negative. We see immediately that this requires:

$$T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tt_\sigma t^\sigma \geq 0 \implies T_{\mu\nu}t^\mu t^\nu \geq \frac{1}{2}Tt_\sigma t^\sigma. \quad (1.210)$$

This is the exact definition of the Strong Energy Condition. Therefore, the SEC is the precise mathematical requirement for gravity to act as an attractive force on all possible timelike trajectories.

Notice that vacuum energy (where $p = -\rho$) violently violates the SEC, because $\rho + 3p = \rho - 3\rho = -2\rho$, which is negative for positive vacuum energy. This SEC violation is the mathematical reason why a positive cosmological constant generates repulsive gravity, leading to the accelerated expansion of the universe.

1.7 The Equivalence Principle Revisited

The Einstein Equivalence Principle (EEP) served as the foundational heuristic for generalizing the laws of Special Relativity to the curved spacetime of General Relativity. By positing that the laws of physics reduce to their Special Relativistic forms in sufficiently small, locally inertial reference frames, we deduced the minimal-coupling procedure: replacing the Minkowski metric $\eta_{\mu\nu}$ with the dynamical metric $g_{\mu\nu}$, and replacing partial derivatives ∂_μ with covariant derivatives ∇_μ .

However, the Equivalence Principle is not a fundamental, mathematically rigorous axiom of the final theory. Rather, it is a phenomenological guide. In practice, the EEP is typically invoked to justify four distinct propositions regarding the formulation of gravitational physics:

1. **General Covariance:** The laws of physics must be expressible in a manifestly coordinate-invariant, tensorial form.
2. **Metric Theory of Gravity:** There exists a symmetric metric tensor $g_{\mu\nu}$ defined on the spacetime manifold, and the curvature of this metric is the sole geometric manifestation of gravity.
3. **Uniqueness of the Gravitational Field:** There do not exist any other long-range fields (e.g., secondary scalar or vector fields) that masquerade as gravity or mediate a "fifth force."
4. **Minimal Coupling:** Matter fields interact with the gravitational field exclusively through the metric and its first derivatives (via the connection). There are no direct algebraic couplings between matter fields and the Riemann curvature tensor.

These four propositions possess different levels of fundamental validity. The first is a mathematical tautology; any physical law can be written in a generally covariant form given sufficient auxiliary fields. The second is the defining hallmark of General Relativity and appears robust. The third is a testable empirical hypothesis. The fourth, however, is merely a low-energy approximation.

1.7.1 The Approximation of Minimal Coupling

From the perspective of modern effective field theory, the minimal-coupling rule is not an absolute law but the leading-order term in a low-energy expansion. Consider a scalar field ϕ with mass m . The minimally coupled action is:

$$S_{\min} = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} m^2 \phi^2 \right]. \quad (1.211)$$

However, the requirements of general covariance and dimensional analysis do not forbid non-minimal couplings. One could easily construct a perfectly valid, generally covariant term that couples the scalar field directly to the Ricci scalar R :

$$S_{\text{non-min}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} \xi R \phi^2 \right], \quad (1.212)$$

where ξ is a dimensionless coupling constant. Such a term vanishes in flat spacetime (where $R = 0$), meaning it satisfies the strict formulation of the Equivalence Principle (that physics reduces to Special Relativity in locally flat regions), yet it violently violates the minimal-coupling heuristic.

In fact, quantum field theory in curved spacetime demonstrates that even if ξ is set to zero classically, loop corrections (vacuum polarization) will dynamically generate this non-minimal coupling term. At macroscopic scales and low energies, the curvature R is incredibly small, rendering these terms physically negligible. Thus, minimal coupling is highly successful in astrophysical applications, but it cannot be viewed as a fundamental geometric prohibition.

1.8 Alternative Theories of Gravity

Given that General Relativity is a classical field theory, and that our motivations for minimal coupling are approximate, it is scientifically imperative to consider alternative formulations of gravity. These modifications are typically motivated by the quest for a quantum theory of gravity, or by the cosmological mysteries of dark matter and dark energy.

1.8.1 Higher-Order Actions: $f(R)$ Gravity

The Einstein-Hilbert action assumes that the gravitational Lagrangian is simply the Ricci scalar R . The most direct generalization is to replace R with an arbitrary analytic function $f(R)$:

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} f(R) + \mathcal{L}_M \right]. \quad (1.213)$$

To determine the modified field equations, we must perform a rigorous variational calculus derivation. We vary the action with respect to the inverse metric $g^{\mu\nu}$:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x [\delta(\sqrt{-g})f(R) + \sqrt{-g}\delta(f(R))] . \quad (1.214)$$

Using the established identity for the variation of the determinant, $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$, and applying the chain rule to the function $f(R)$, we obtain:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[-\frac{1}{2}f(R)g_{\mu\nu}\delta g^{\mu\nu} + f'(R)\delta R \right] , \quad (1.215)$$

where $f'(R) \equiv df/dR$. We must now expand the variation of the Ricci scalar, $\delta R = \delta(g^{\mu\nu}R_{\mu\nu})$:

$$\delta R = R_{\mu\nu}\delta g^{\mu\nu} + g^{\mu\nu}\delta R_{\mu\nu} . \quad (1.216)$$

Recall the Palatini identity for the variation of the Ricci tensor:

$$\delta R_{\mu\nu} = \nabla_\lambda(\delta\Gamma_{\mu\nu}^\lambda) - \nabla_\nu(\delta\Gamma_{\mu\lambda}^\lambda) . \quad (1.217)$$

By substituting the variation of the Levi-Civita connection in terms of the metric variation, one finds that the contraction $g^{\mu\nu}\delta R_{\mu\nu}$ can be expressed as a total derivative involving covariant d'Alembertians:

$$g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\mu \nabla_\nu (\delta g^{\mu\nu}) - \square(g_{\mu\nu}\delta g^{\mu\nu}) , \quad (1.218)$$

where $\square = \nabla^\sigma \nabla_\sigma$. We substitute this into our action variation:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} \right] \delta g^{\mu\nu} + \frac{1}{16\pi G} \int d^4x \sqrt{-g} f'(R) [\nabla_\mu \nabla_\nu - g_{\mu\nu}\square] \delta g^{\mu\nu} . \quad (1.219)$$

The second integral contains derivatives acting on $\delta g^{\mu\nu}$. We must integrate by parts twice to shift the derivatives onto $f'(R)$. Assuming the boundary terms vanish at infinity, the integration by parts yields:

$$\int d^4x \sqrt{-g} f'(R) \nabla_\mu \nabla_\nu (\delta g^{\mu\nu}) = \int d^4x \sqrt{-g} [\nabla_\mu \nabla_\nu f'(R)] \delta g^{\mu\nu} , \quad (1.220)$$

and similarly for the d'Alembertian term. Gathering all terms proportional to $\delta g^{\mu\nu}$:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} + \nabla_\mu \nabla_\nu f'(R) - g_{\mu\nu}\square f'(R) \right] \delta g^{\mu\nu} . \quad (1.221)$$

Adding the variation of the matter action, defined as $\delta S_M = -\frac{1}{2} \int d^4x \sqrt{-g} T_{\mu\nu} \delta g^{\mu\nu}$, and demanding that the total variation vanishes yields the exact **modified field equations for $f(R)$ gravity**:

$$f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} + (\nabla_\mu \nabla_\nu - g_{\mu\nu}\square) f'(R) = 8\pi G T_{\mu\nu} . \quad (1.222)$$

Physical Interpretation: If $f(R) = R$, then $f'(R) = 1$. The derivative terms $\nabla_\mu \nabla_\nu (1)$ trivially vanish, and equation (1.222) perfectly recovers standard General Relativity. However, for non-linear functions (e.g., $f(R) = R + \alpha R^2$), the field equations become fourth-order differential equations for the metric, due to the $\square f'(R)$ term containing second derivatives of the Ricci scalar (which itself contains second derivatives of the metric). Such theories introduce a new, propagating scalar degree of freedom embedded within the geometry.

1.8.2 Scalar-Tensor Theories

Another prominent class of alternatives abandons the idea that geometry is described solely by a tensor field. In Scalar-Tensor theories, such as Brans-Dicke theory, the gravitational coupling constant G is promoted to a dynamical scalar field $\phi(x)$.

The action for a generalized scalar-tensor theory takes the form:

$$S = \int d^4x \sqrt{-g} \left[\phi R - \frac{\omega(\phi)}{\phi} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) + \mathcal{L}_M \right], \quad (1.223)$$

where $\omega(\phi)$ is a coupling parameter and $V(\phi)$ is a potential. Unlike the non-minimal coupling in equation (1.212), here the scalar field dictates the absolute strength of gravity. Varying this action with respect to $g^{\mu\nu}$ generates a modified Einstein equation where the scalar field acts as a geometric source term. Varying the action with respect to ϕ generates a generalized Klein-Gordon equation, demonstrating that the matter distribution $T_{\mu\nu}$ acts as a direct source for the scalar field. This violates the Strong Equivalence Principle, as the local gravitational "constant" becomes dependent on the surrounding cosmic matter distribution.

1.8.3 Torsion and Non-Christoffel Connections

General Relativity rests upon the Fundamental Theorem of Riemannian Geometry, which assumes the connection $\Gamma_{\mu\nu}^\rho$ is both metric-compatible ($\nabla_\lambda g_{\mu\nu} = 0$) and symmetric in its lower indices ($\Gamma_{\mu\nu}^\rho = \Gamma_{\nu\mu}^\rho$).

If we relax the symmetry requirement, the antisymmetric part of the connection defines a new valid tensor, the **torsion tensor**:

$$T_{\mu\nu}^\rho = \Gamma_{\mu\nu}^\rho - \Gamma_{\nu\mu}^\rho. \quad (1.224)$$

In Einstein-Cartan theory, spacetime is permitted to have non-vanishing torsion. While mass and energy source spacetime curvature, intrinsic quantum mechanical spin is postulated to source spacetime torsion. Because macroscopic objects have randomly oriented intrinsic spins that average to zero, torsion is utterly negligible in the Solar System. However, at extreme densities (such as inside a collapsing neutron star), torsion would manifest as a repulsive spin-spin interaction, potentially halting the formation of a physical singularity.

1.8.4 Additional Tensor Fields and Geometric Generalizations

When exploring alternatives to General Relativity, one might consider relaxing the fundamental geometric axioms of the manifold, such as metric compatibility. The Fundamental Theorem of Riemannian Geometry guarantees a unique Levi-Civita connection $\tilde{\Gamma}_{\mu\nu}^\rho$ that is both symmetric (torsion-free) and metric-compatible ($\tilde{\nabla}_\lambda g_{\mu\nu} = 0$).

If we drop the requirement of metric compatibility, the covariant derivative of the metric no longer vanishes ($\nabla_\lambda g_{\mu\nu} = Q_{\lambda\mu\nu} \neq 0$), defining the **non-metricity tensor** $Q_{\lambda\mu\nu}$. However, any arbitrary affine connection $\Gamma_{\mu\nu}^\rho$ can be strictly decomposed into the unique Levi-Civita connection plus a tensorial correction $C_{\mu\nu}^\rho$ (often related to the contortion or defect tensor):

$$\Gamma_{\mu\nu}^\rho = \tilde{\Gamma}_{\mu\nu}^\rho + C_{\mu\nu}^\rho. \quad (1.225)$$

Because the difference between any two connections is a true tensor, $C_{\mu\nu}^\rho$ transforms covariantly. Consequently, substituting this decomposed connection into the Riemann curvature tensor effectively splits the curvature into a purely Riemannian part (calculated solely from $g_{\mu\nu}$) plus covariant terms involving $C_{\mu\nu}^\rho$ and its derivatives.

Physical Interpretation: From the perspective of the action principle, a theory of gravity formulated with a non-metric-compatible connection (or a connection with torsion) is mathematically isomorphic to standard General Relativity (using the Levi-Civita connection) coupled to additional, independent tensor fields. Because these extra degrees of freedom do not fundamentally alter the underlying Riemannian structure, theories involving non-metricity or torsion are often viewed not as radical replacements for General Relativity, but simply as General Relativity augmented by exotic matter fields.

1.9 Variational Derivations of Advanced Field Equations

To solidify the Lagrangian formulation of General Relativity, we present explicitly expanded derivations of two critical results: the derivation of the electromagnetic energy-momentum tensor via metric variation, and the Palatini formulation of gravity, which dynamically generates metric compatibility rather than assuming it a priori.

1.9.1 The Electromagnetic Stress-Energy Tensor

Consider the action for the electromagnetic field in a curved spacetime background. The Maxwell Lagrangian density is constructed from the antisymmetric Faraday tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, where A_μ is the four-potential. The action is:

$$S_{\text{EM}} = \int d^4x \sqrt{-g} \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + A_\mu J^\mu \right), \quad (1.226)$$

where J^μ is the conserved charge current. To derive the stress-energy tensor $T_{\mu\nu}$ using the Hilbert definition, we must vary this action with respect to the inverse metric $g^{\rho\sigma}$.

First, we rewrite the scalar contraction $F^{\mu\nu} F_{\mu\nu}$ explicitly in terms of the inverse metric to expose all metric dependencies:

$$F^{\mu\nu} F_{\mu\nu} = g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta}. \quad (1.227)$$

Crucially, the Faraday tensor $F_{\mu\nu}$ is defined purely by partial derivatives of the potential A_μ , which are entirely independent of the metric. Thus, $\delta F_{\mu\nu} = 0$.

We vary the action S_{EM} with respect to $g^{\rho\sigma}$:

$$\begin{aligned} \delta S_{\text{EM}} &= \delta \int d^4x \left[\sqrt{-g} \left(-\frac{1}{4} g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} \right) \right] \\ &= \int d^4x \left[(\delta \sqrt{-g}) \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \right) - \frac{1}{4} \sqrt{-g} \delta (g^{\mu\alpha} g^{\nu\beta}) F_{\mu\nu} F_{\alpha\beta} \right]. \end{aligned} \quad (1.228)$$

Recall the identity for the variation of the metric determinant: $\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} g_{\rho\sigma} \delta g^{\rho\sigma}$.

Next, we expand the variation of the metric product using the product rule:

$$\delta (g^{\mu\alpha} g^{\nu\beta}) = (\delta g^{\mu\alpha}) g^{\nu\beta} + g^{\mu\alpha} (\delta g^{\nu\beta}). \quad (1.229)$$

Substituting this into the second term of the integral:

$$-\frac{1}{4} \sqrt{-g} \left[(\delta g^{\mu\alpha}) g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} + g^{\mu\alpha} (\delta g^{\nu\beta}) F_{\mu\nu} F_{\alpha\beta} \right]. \quad (1.230)$$

We relabel dummy indices to factor out $\delta g^{\rho\sigma}$. In the first term, let $\mu \rightarrow \rho$ and $\alpha \rightarrow \sigma$. In the second term, let $\nu \rightarrow \rho$ and $\beta \rightarrow \sigma$:

$$-\frac{1}{4} \sqrt{-g} \left[\delta g^{\rho\sigma} (g^{\nu\beta} F_{\rho\nu} F_{\sigma\beta}) + \delta g^{\rho\sigma} (g^{\mu\alpha} F_{\mu\rho} F_{\alpha\sigma}) \right]. \quad (1.231)$$

Using the antisymmetry of the Faraday tensor ($F_{\mu\rho} = -F_{\rho\mu}$ and $F_{\alpha\sigma} = -F_{\sigma\alpha}$), the second term becomes identical to the first:

$$-\frac{1}{4} \sqrt{-g} \left[2 \delta g^{\rho\sigma} F_\rho{}^\beta F_{\sigma\beta} \right] = -\frac{1}{2} \sqrt{-g} F_{\rho\lambda} F_\sigma{}^\lambda \delta g^{\rho\sigma}. \quad (1.232)$$

Combining both parts of the variation:

$$\delta S_{\text{EM}} = \int d^4x \sqrt{-g} \left[\frac{1}{8} g_{\rho\sigma} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} F_{\rho\lambda} F_\sigma{}^\lambda \right] \delta g^{\rho\sigma}. \quad (1.233)$$

Using the Hilbert definition of the stress-energy tensor, $T_{\rho\sigma} = \frac{-2}{\sqrt{-g}} \frac{\delta S_{\text{EM}}}{\delta g^{\rho\sigma}}$, we multiply the bracketed term by -2 to obtain the final result:

$$T_{\rho\sigma} = F_{\rho\lambda} F_\sigma{}^\lambda - \frac{1}{4} g_{\rho\sigma} F_{\mu\nu} F^{\mu\nu}. \quad (1.234)$$

Properties: By contracting with the inverse metric ($T = g^{\rho\sigma}T_{\rho\sigma}$), we find $T = F^{\sigma\lambda}F_{\sigma\lambda} - \frac{1}{4}(4)F_{\mu\nu}F^{\mu\nu} = 0$. The electromagnetic stress-energy tensor is strictly traceless in four dimensions, reflecting the conformal invariance of Maxwell's equations and the masslessness of the photon.

1.9.2 The Palatini Formulation of General Relativity

In the standard derivation of Einstein's field equations from the Einstein-Hilbert action, we assumed from the outset that the connection $\Gamma_{\mu\nu}^\lambda$ was the Levi-Civita connection, fully determined by the metric. The Palatini formulation takes a more fundamental approach: it treats the metric $g_{\mu\nu}$ and the connection $\Gamma_{\mu\nu}^\lambda$ as entirely independent dynamical variables.

The action is formally identical, but the Ricci tensor is now considered a function solely of the connection, $R_{\mu\nu}(\Gamma)$:

$$S[g, \Gamma] = \int d^4x \sqrt{-g} g^{\mu\nu} R_{\mu\nu}(\Gamma). \quad (1.235)$$

Varying the action with respect to the metric $g^{\mu\nu}$ proceeds exactly as before, trivially yielding $R_{\mu\nu}(\Gamma) - \frac{1}{2}R(\Gamma)g_{\mu\nu} = 0$.

The profoundly new physics emerges when we vary the action with respect to the connection. Because $\delta\Gamma_{\mu\nu}^\lambda$ is a tensor, the variation of the Ricci tensor is given by the Palatini identity:

$$\delta R_{\mu\nu} = \nabla_\lambda(\delta\Gamma_{\mu\nu}^\lambda) - \nabla_\nu(\delta\Gamma_{\mu\lambda}^\lambda). \quad (1.236)$$

We insert this into the action variation:

$$\delta_\Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} [\nabla_\lambda(\delta\Gamma_{\mu\nu}^\lambda) - \nabla_\nu(\delta\Gamma_{\mu\lambda}^\lambda)] = 0. \quad (1.237)$$

Because the connection is an independent variable, we *cannot* assume that $\nabla_\lambda g^{\mu\nu} = 0$. Therefore, we cannot pull $g^{\mu\nu}$ inside the covariant derivatives. Instead, we must integrate by parts. The generalized integration by parts rule in curved space (derived from the divergence theorem) states that for a vector V^λ and scalar f :

$$\int d^4x \sqrt{-g} f \nabla_\lambda V^\lambda = - \int d^4x \sqrt{-g} V^\lambda \nabla_\lambda f + \text{boundary terms}. \quad (1.238)$$

However, the object we are varying against is a tensor density, $\sqrt{-g}g^{\mu\nu}$. Applying the chain rule in reverse (integration by parts) to our variation yields:

$$\delta_\Gamma S = \int d^4x \left[-\nabla_\lambda(\sqrt{-g}g^{\mu\nu})\delta\Gamma_{\mu\nu}^\lambda + \nabla_\nu(\sqrt{-g}g^{\mu\nu})\delta\Gamma_{\mu\lambda}^\lambda \right]. \quad (1.239)$$

We relabel dummy indices in the second term to factor out $\delta\Gamma_{\mu\nu}^\lambda$. Let $\nu \rightarrow \lambda$ and $\lambda \rightarrow \nu$:

$$\delta_\Gamma S = \int d^4x \left[-\nabla_\lambda(\sqrt{-g}g^{\mu\nu}) + \delta_\lambda^\nu \nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) \right] \delta\Gamma_{\mu\nu}^\lambda = 0. \quad (1.240)$$

For this action to be stationary under arbitrary variations $\delta\Gamma_{\mu\nu}^\lambda$, the quantity in the brackets must vanish identically:

$$\nabla_\lambda(\sqrt{-g}g^{\mu\nu}) - \delta_\lambda^\nu \nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0. \quad (1.241)$$

We can solve this differential equation by contracting indices. Taking the trace by setting $\nu = \lambda$ yields:

$$\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) - 4\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0 \implies -3\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0. \quad (1.242)$$

This requires that $\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0$. Substituting this back into equation (1.241) eliminates the second term entirely, leaving the strict requirement:

$$\nabla_\lambda(\sqrt{-g}g^{\mu\nu}) = 0. \quad (1.243)$$

Expanding the covariant derivative of the scalar density $\sqrt{-g}$:

$$\sqrt{-g}\nabla_\lambda g^{\mu\nu} + g^{\mu\nu}\nabla_\lambda \sqrt{-g} = 0. \quad (1.244)$$

By utilizing the identity $\nabla_\lambda \sqrt{-g} = \frac{1}{2}\sqrt{-g}g^{\alpha\beta}\nabla_\lambda g_{\alpha\beta}$, one can algebraically show that this implies the covariant derivative of the metric itself must vanish:

$$\nabla_\lambda g^{\mu\nu} = 0. \quad (1.245)$$

Geometric Conclusion: The Palatini formulation is extraordinarily profound. By treating the metric and the connection as completely independent geometric objects, the principle of least action dynamically forces the connection to be metric-compatible. We do not have to assume Riemannian geometry a priori; the dynamics of the gravitational field independently demand it.